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&EDB				11				Information-sheet																19.06.2018				PIE/PLE																							
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&EFS				13				Control voltage								=0				+MCC				19.06.2018				PIE/PLE																							
&EFS				14				Control voltage AC								=0				+MCC				19.06.2018				PIE/PLE																							
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&EFS				16				Cabinet fan								=0				+MCC				19.06.2018				PIE/PLE																							
&EFS				17				Emergency stop								=0				+MCC				19.06.2018				PIE/PLE																							
&EFS				18				Separator								=_1_001				+MCC				19.06.2018				PIE/PLE																							
&EFS				19				Separator								=_1_001				+MCC				19.06.2018				PIE/PLE																							
				Date				19.06.2018				Kunde: -				GEA Westfalia Separator Group GmbH								Benennung/Description: Table of contents								GEA IO 4				=															
				Editor				HAN/PLE				Customer:																												+											
				Tested				19.06.2018/GR				Projekt:																												Project:											
Changes				Date				Name								© This document is COPYRIGHTED by GEA Westfalia Separator Group GmbH. All rights being reserved.												Sprache:				Zeichnungs Nr.:				DCC				Bl./Sh.											
																												Language:				en				Drawing No.:				8552-7300-086				& EAB				104 Bl./Sh.			

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&EFS	39	PLC analog earth bridges S7-1200	=00	+PLC	19.06.2018	PIE/PLE
&EFS	40	PLC analog earth bridges S7-1200	=00	+PLC	19.06.2018	PIE/PLE
&EFS	41	Operating panel KTP400	=00	+PLC	19.06.2018	PIE/PLE
&EFS	42	Power supply (S7-1200)	=00	+PLC	19.06.2018	PIE/PLE
&EFS	43	PLC overview Onboard S7-1200	=00	+PLC	19.06.2018	PIE/PLE
&EFS	44	PLC overview S7-1200 4AI/2AO	=00	+PLC	19.06.2018	PIE/PLE
&EFS	45	PLC overview S7-1200 16DI	=00	+PLC	19.06.2018	PIE/PLE
&EFS	46	PLC overview S7-1200 16DO	=00	+PLC	19.06.2018	PIE/PLE
&EFS	47	PLC overview S7-1200 4AI/2AO	=00	+PLC	19.06.2018	PIE/PLE
&ELU	48	Arrangement drawing		+MCC	19.06.2018	PIE/PLE
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&ELU	50	Arrangement drawing		+PLC	19.06.2018	PIE/PLE
&ELU	51	Arrangement drawing		+PLC	19.06.2018	PIE/PLE
&EMA	52	Terminal connection diagram =0+MCC-N		+MCC	19.06.2018	PIE/PLE
&EMA	53	Terminal connection diagram =0+MCC-PE		+MCC	19.06.2018	PIE/PLE
&EMA	54	Terminal connection diagram =0+MCC-X1		+MCC	19.06.2018	PIE/PLE
&EMA	55	Terminal connection diagram =0+MCC-X1.1		+MCC	19.06.2018	PIE/PLE
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&EMA	61	Terminal connection diagram =0+MCC-X2000		+MCC	19.06.2018	PIE/PLE
&EMA	62	Terminal connection diagram =0+MCC-XQ1		+MCC	19.06.2018	PIE/PLE
&EMA	63	Terminal connection diagram =_1_001+MCC-XT1		+MCC	19.06.2018	PIE/PLE
&EMA	64	Terminal connection diagram =_8+MCC-X1		+MCC	19.06.2018	PIE/PLE
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&EMA	66	Terminal connection diagram =SEP+MCC-X11.1		+MCC	19.06.2018	PIE/PLE
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&EMA	76	Terminal connection diagram =_9_034+PLC-X2		+PLC	19.06.2018	PIE/PLE

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&EMA	78	Terminal connection diagram =_9_550+PLC-X2		+PLC	19.06.2018	PIE/PLE
&EMA	79	Terminal connection diagram =_7_137+PLC-X2		+PLC	19.06.2018	PIE/PLE
&EMA	80	Terminal connection diagram =_7_131+PLC-X2		+PLC	19.06.2018	PIE/PLE
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&EMA	86	Terminal connection diagram =SEP+JB-X10		+JB	19.06.2018	PIE/PLE
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&EMB	93	Cable layout			19.06.2018	PIE/PLE
&EMB	94	Cable layout			19.06.2018	PIE/PLE
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&EMB	97	Potential equalization separator			19.06.2018	PIE/PLE
&EPB	98	Parts list			19.06.2018	PIE/PLE
&EPB	99	Parts list			19.06.2018	PIE/PLE
&EPB	100	Parts list			19.06.2018	PIE/PLE
&EPB	101	Parts list			19.06.2018	PIE/PLE
&EPB	102	Parts list			19.06.2018	PIE/PLE
&EPB	103	Parts list			19.06.2018	PIE/PLE
&EPB	104	Parts list			19.06.2018	PIE/PLE

Revision overview

Revision name	Revision comment	Revision on the basis of change	Page name	Author	Date

Overview structure indicator

Complete name	Labelling	Structure descprition	Complete name	Labelling	Structure descprition
&EAA	Document type	Cover sheet	=_9_123	Plant	Flow
&EAB	Document type	Table of contents	=_7_024	Plant	Valve
&EBH	Document type	Revision overview	=_7_025	Plant	Valve
&EDB	Document type	Structure indentification overview / instructions	=_7_022	Plant	Valve
&EFS	Document type	Circuit diagram	=_7_023	Plant	Valve
&ELU	Document type	Arrangement drawing	=_7_106	Plant	Valve
&EMA	Document type	Terminal connection diagram, terminal connecting plan	=_7_034/Y1	Plant	Valve
&EMB	Document type	Cable layout	=_7_032	Plant	Valve
&EPB	Document type	Article parts list	=_9_500	Plant	Oil unit
=CUSTOMER	Plant	Customer	=_9_018	Plant	Vibro
=0	Plant	General	=100	Plant	Signal exchange
=_1_001	Plant	Separator	=00	Plant	PLC
=_3_002	Plant	Drive	=SEP	Plant	Separator
=_8	Plant	Sight glass illumination	+MCC	Mounting place	Motor starter
=_8_005	Plant	Sight glass illumination	+PLC	Mounting place	Control cabinet
=_8_006	Plant	Sight glass illumination	+JB	Mounting place	Terminal box
=_9_034	Plant	Control air	+L	Mounting place	Locally at the separator
=Y	Plant	Valves			
=_9_550	Plant	Discharge pressure			
=_7_137	Plant	Control valve			
=_7_131	Plant	Centripetal pump adjustment			
=_9_022	Plant	Speed bowl			
=_9_019	Plant	Operating water			

Instruction / specification

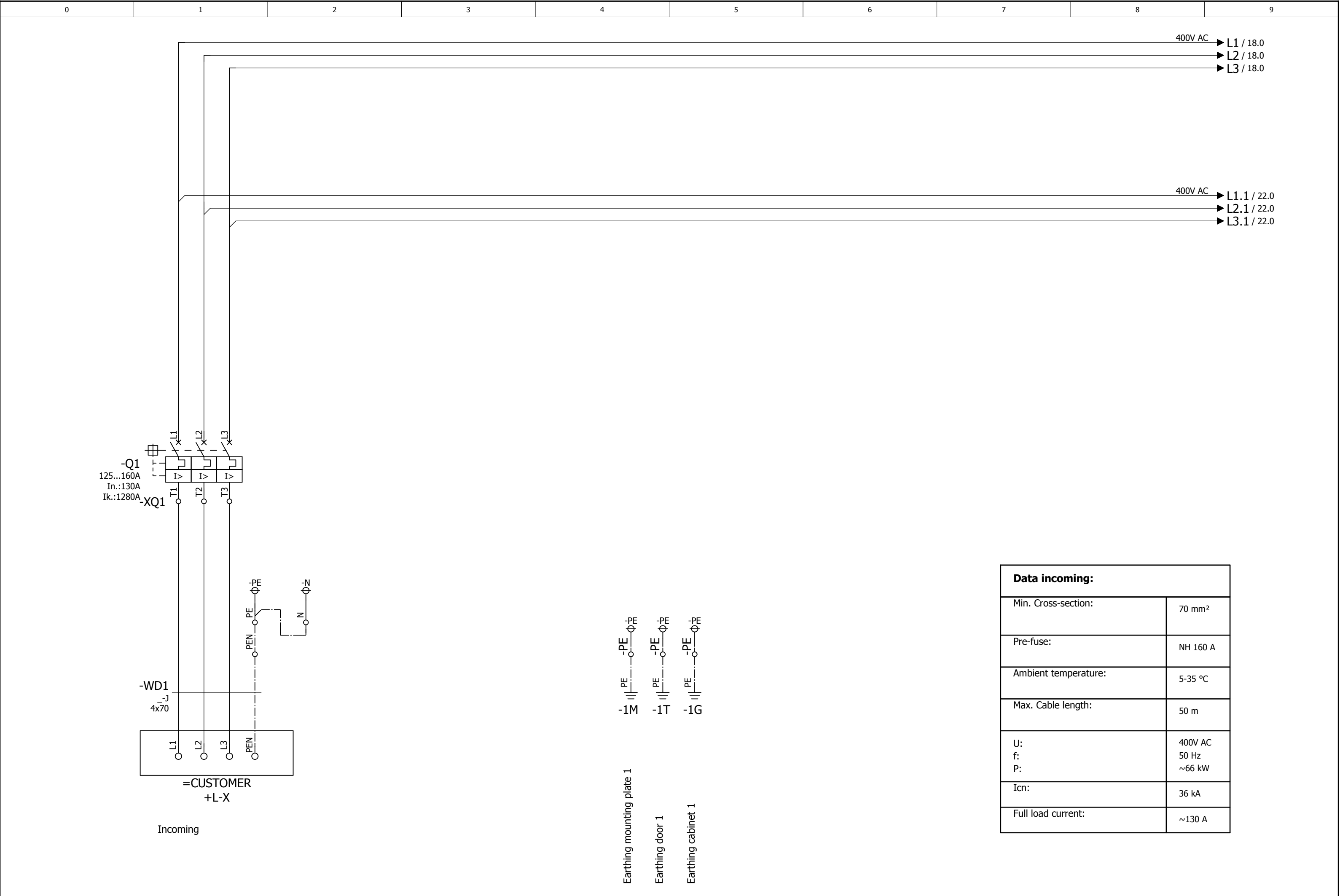
Control cabinet :		
	Denomination control cabinet	MCC
	Painting cabinet	RAL 7035
	Protective rating	IP54
	Ambient temperature	5-35 °C
Control cabinet:		
	Denomination control cabinet	PLC
	Painting cabinet	Niro stainless steel 1.4301
	Protective rating	IP54
	Ambient temperature	5-35 °C
Control cabinet:		
	Denomination control cabinet	-
	Painting cabinet	-
	Protective rating	-
	Ambient temperature	-

Version:		
	Core indentification	No
	Potential designation	No
	Halogen free wiring / channels	No
	Cable marking plates	No
	Classification	No
	Extra specification	No
	Labels	No
Interface:		
	Bus connection to customer PLC	Profibus DP
Incoming:		
	Mains-/ input voltage	400V AC
	Mains frequency	50 Hz
	Mains form	TN-C-S
	Rotary field	Clockwise rotary field
	Voltage tolerance	+/-5 %
	Cable inlet	Bottom
	Incoming redundant	No
Control voltages:		
	Control voltage 1	230V AC
	Control voltage 2	24V DC

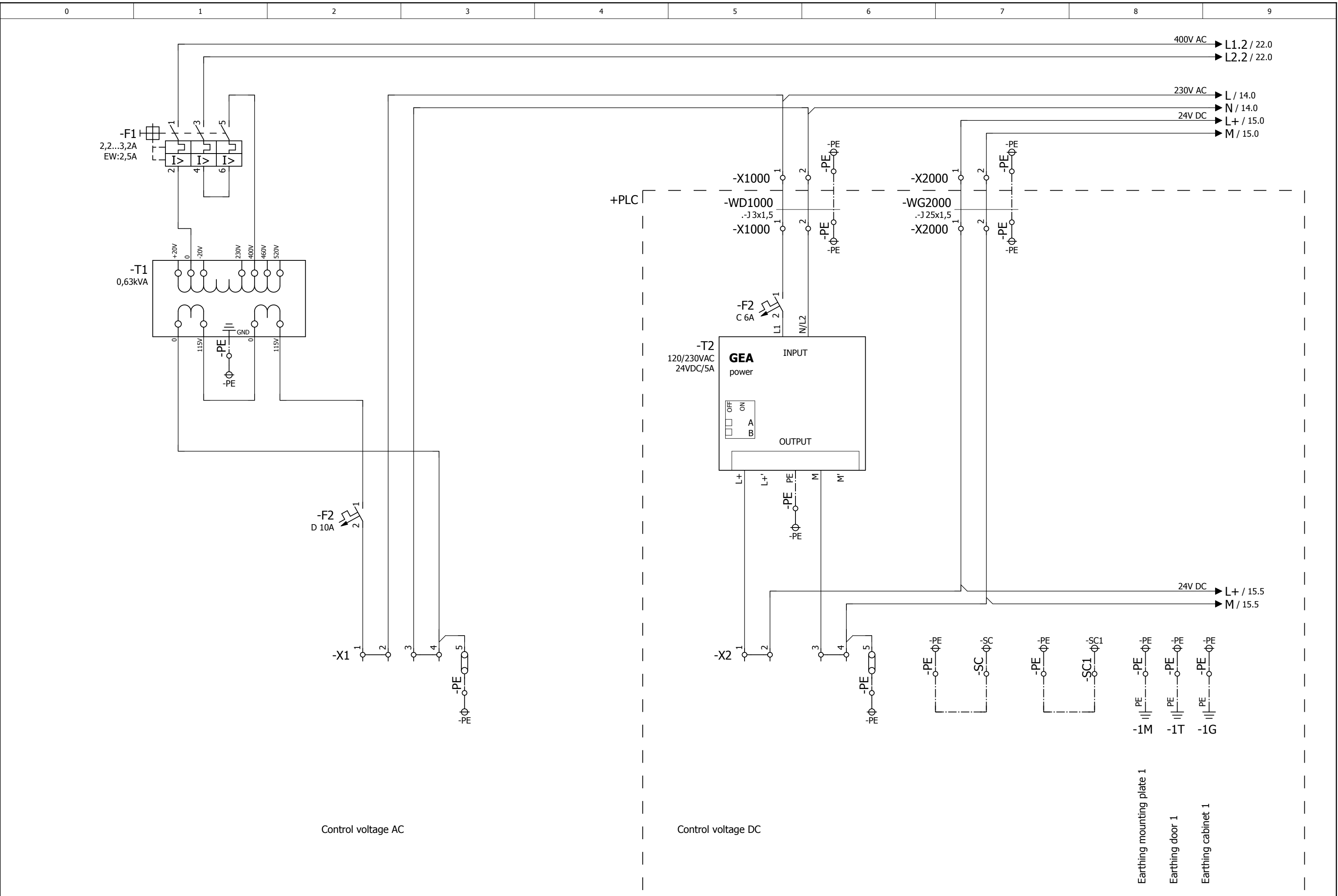
Wiring

Electrical circuits	Conductor	Colour	Cross-section
Main circuits400V AC	L1		≥ 1,5mm²
	L2		"
	L3		"
	N		≥ 1,5mm²
	PE		≥ 2,5mm²

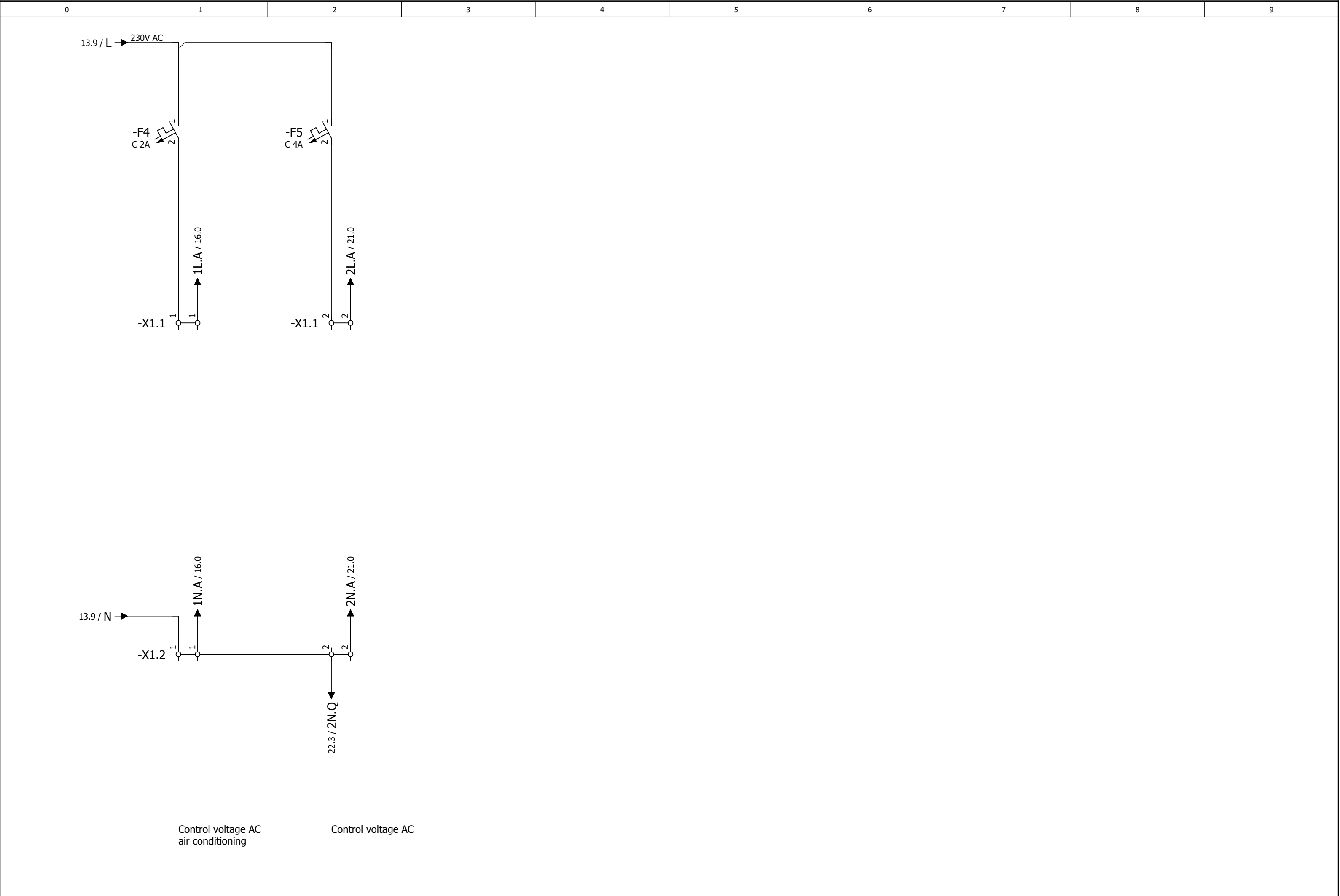
Electrical circuits	Conductor	Colour	Cross-section
Control voltage230V AC	L	Red	≥ 0,75mm²
	N	Red / white	"
Control voltage24V AC	L	Red	≥ 0,75mm²
	N	Red / white	"
Control voltage24V DC	+	Blue	≥ 0,75mm²
	-	Blue / white	"
Analog screened	L+	White	≥ 0,75mm²
	M	Brown	"
Current transformer lines		Black	≥ 2,5mm²
Conductors with external voltage (AC/DC)		Orange	≥ 1,0mm²



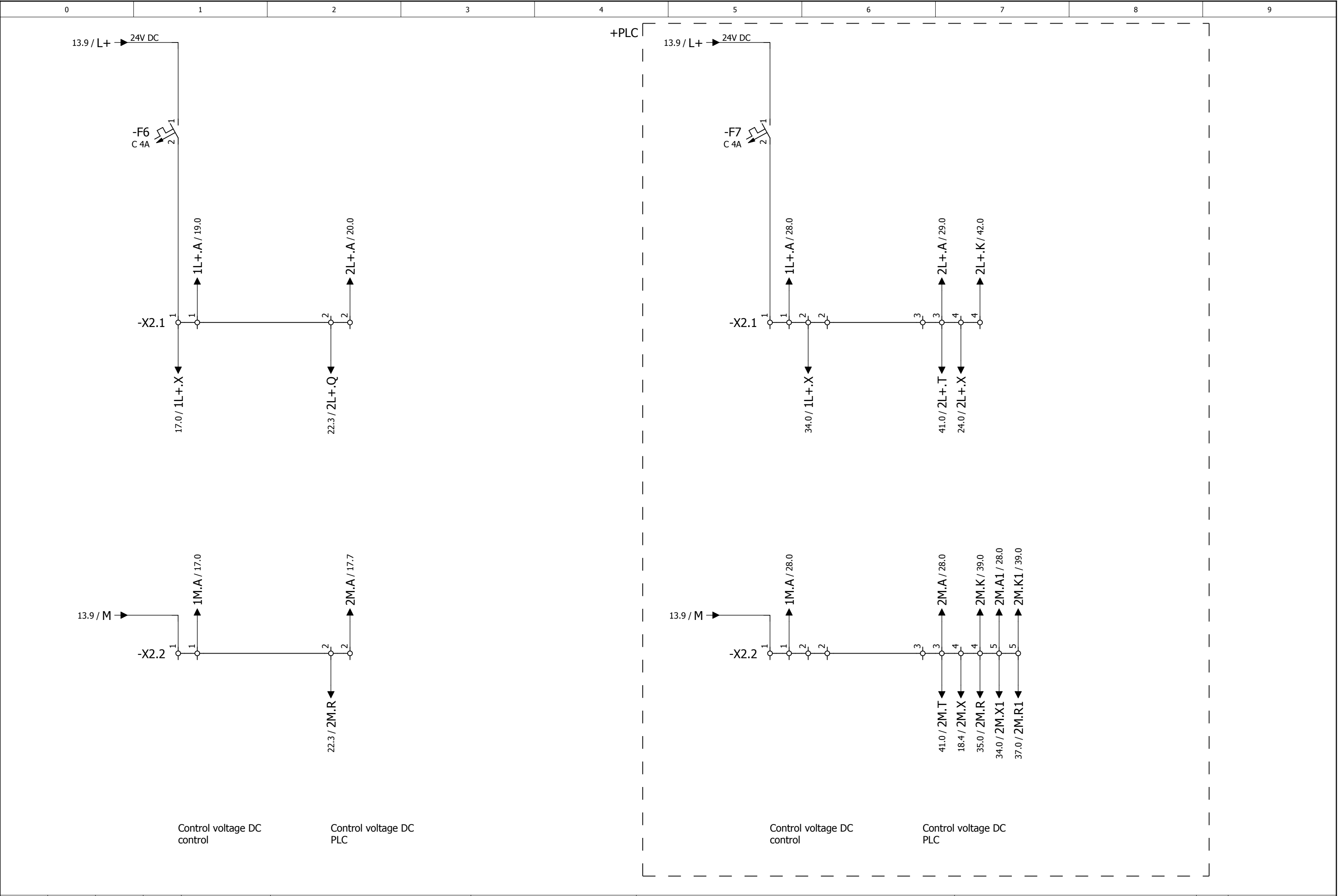
Data incoming:	
Min. Cross-section:	70 mm²
Pre-fuse:	NH 160 A
Ambient temperature:	5-35 °C
Max. Cable length:	50 m
U: f: P:	400V AC 50 Hz ~66 kW
Icn:	36 kA
Full load current:	~130 A

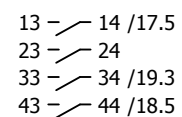


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			Editor	HAN/PLE					+ MCC					
			Tested	19.06.2018/GR				Projekt: Project:						
Changes	Date	Name			© This document is COPYRIGHTED by GEA Westfalia Separator Group GmbH. All rights being reserved.				Sprache: Language:	en	Zeichnungs Nr.: Drawing No.:	8552-7300-086	DCC & EFS	Bl./Sh. 13 104 Bl./Sh.



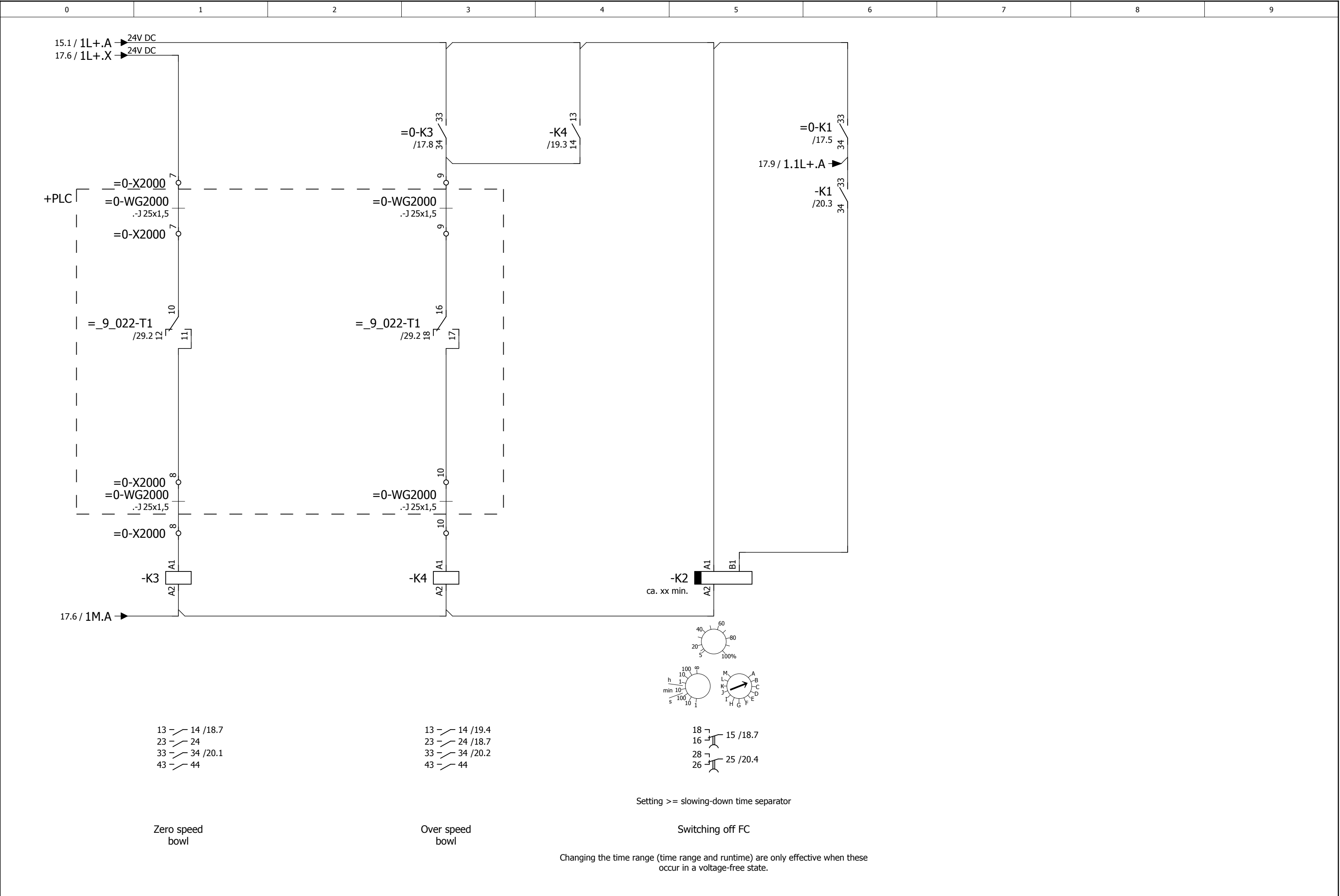
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			Tested	19.06.2018/GR						Sprache: Language:	en	Zeichnungs Nr.: Drawing No.:	8552-7300-086
Changes	Date	Name			© This document is COPYRIGHTED by GEA Westfalia Separator Group GmbH. All rights being reserved.								



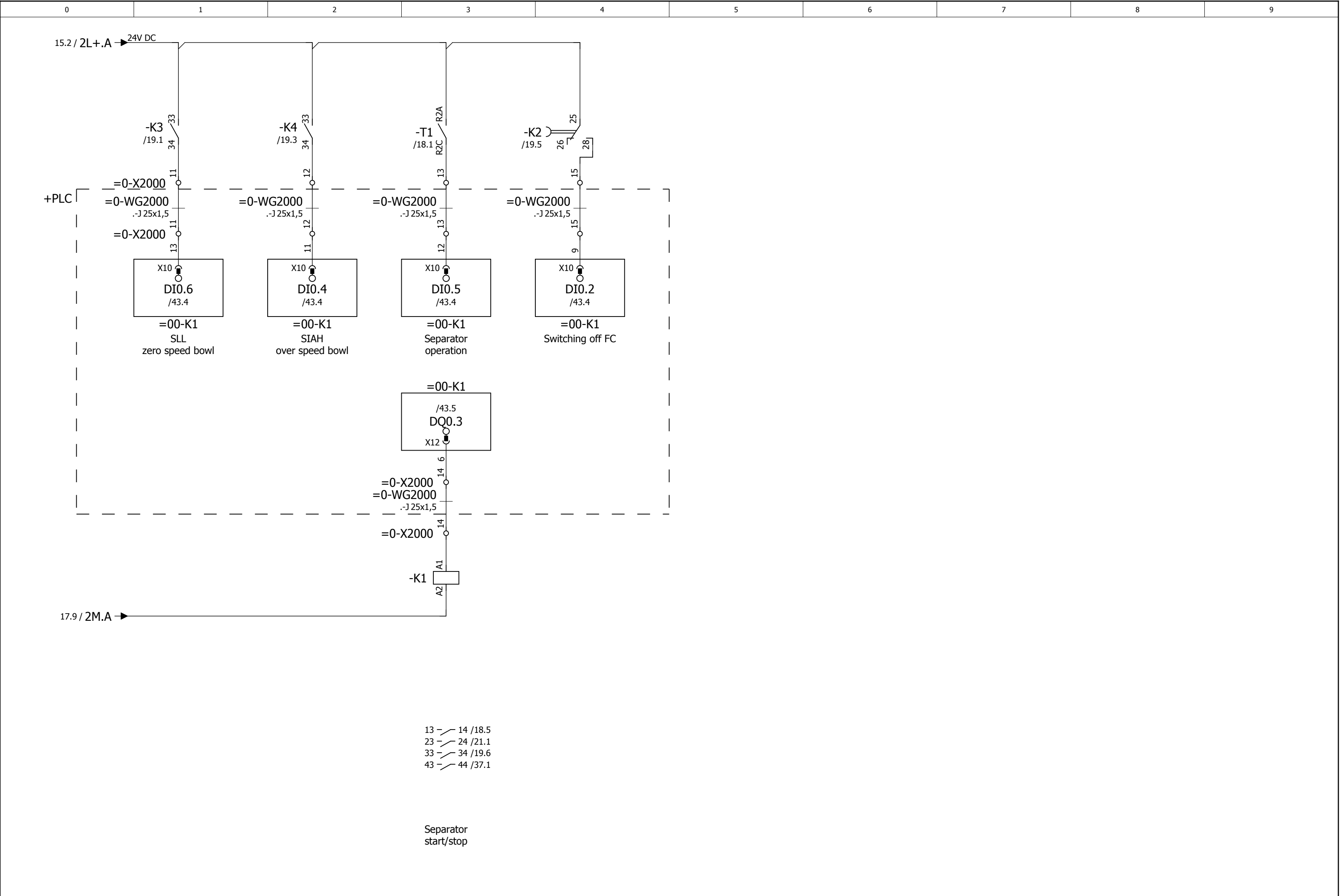


Emergency stop reset

		Date	19.06.2018	Kunde: -	 GEA Westfalia Separator Group GmbH	Benennung/Description: Emergency stop	GEA IO 4			= 0		
		Editor	HAN/PLE	Customer:			+ MCC					
		Tested	19.06.2018/GR	Projekt: Project:								
Changes	Date	Name	© This document is COPYRIGHTED by GEA Westfalia Separator Group GmbH. All rights being reserved.				Sprache: Language:	en	Zeichnungs Nr.: Drawing No.:	8552-7300-086	DCC & EFS	Bl./Sh. 17 104 Bl./Sh.



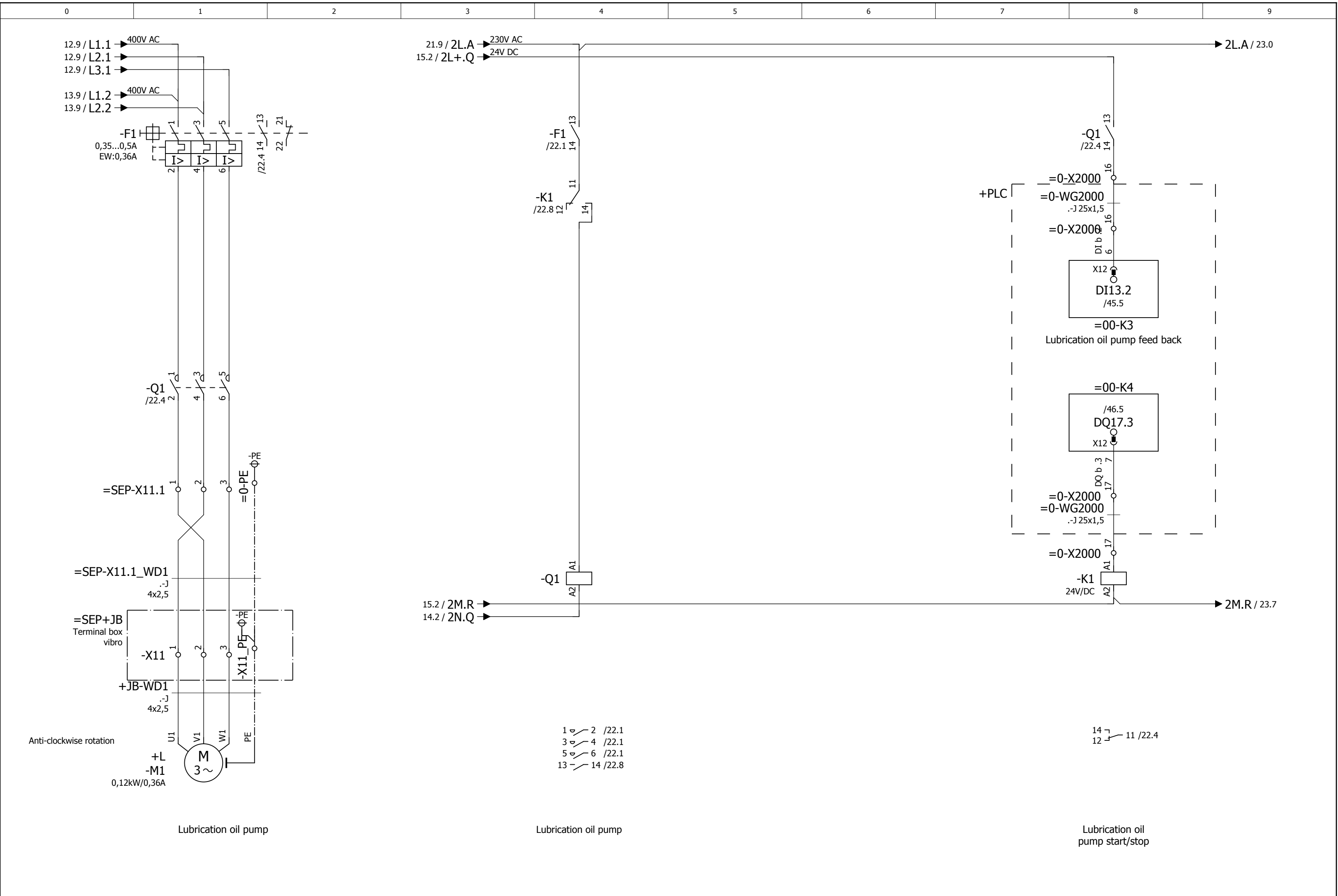
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			Editor	HAN/PLE	Customer:					+ MCC			
			Tested	19.06.2018/GR	Projekt: Project:			Sprache: Language:		Zeichnungs Nr.: Drawing No.:		DCC & EFS	Bl./Sh. 19 104 Bl./Sh.
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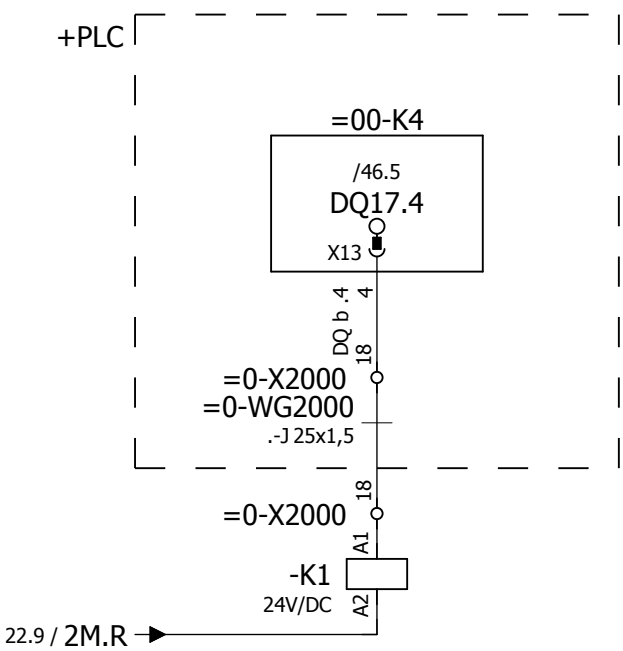
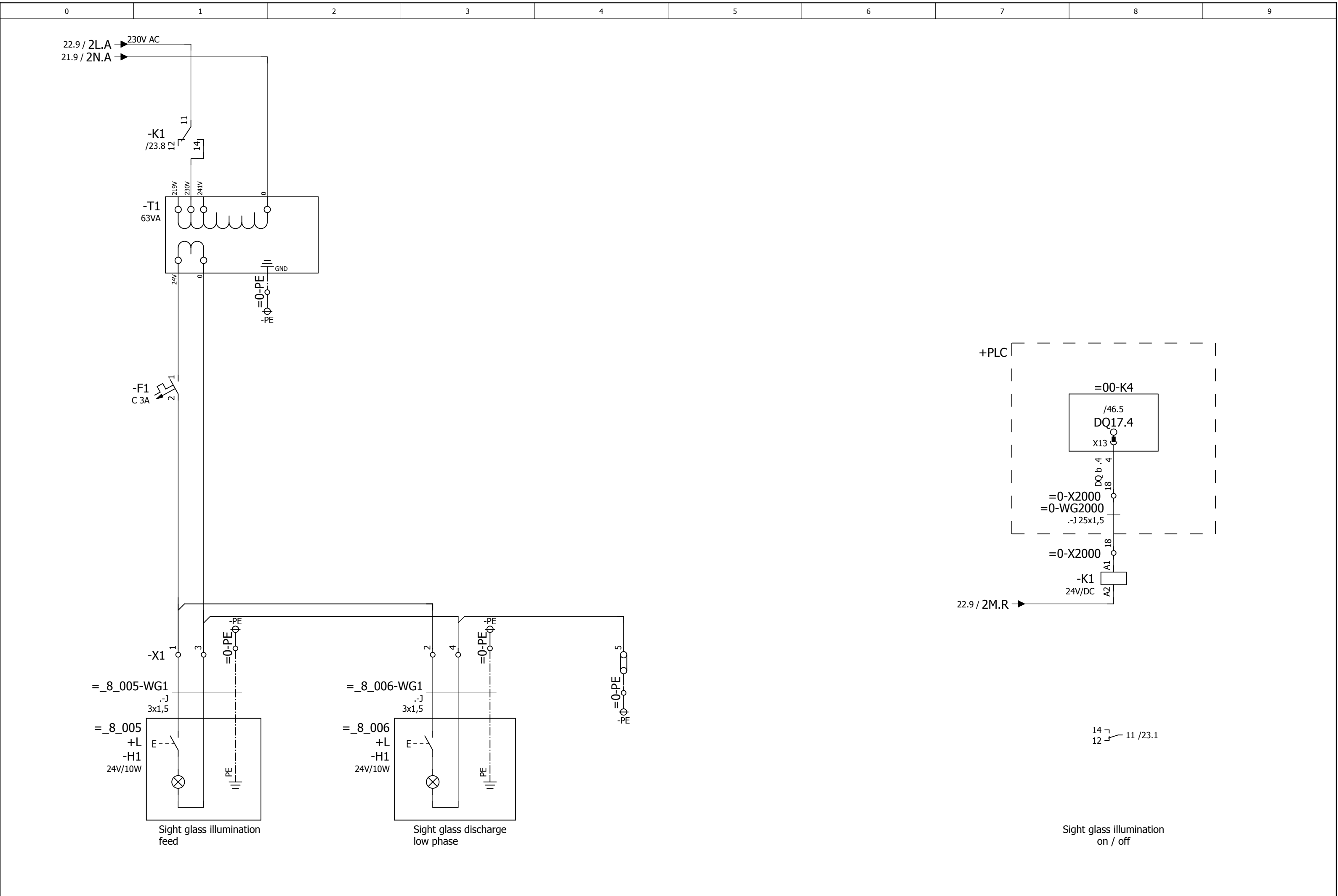
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			Editor	HAN/PLE	Customer:						+ MCC		
			Tested	19.06.2018/GR	Projekt: Project:								
Changes	Date	Name			© This document is COPYRIGHTED by GEA Westfalia Separator Group GmbH. All rights being reserved.				Sprache: Language:	en	Zeichnungs Nr.: Drawing No.:	8552-7300-086	DCC &EFS



			Date	19.06.2018	Kunde: - Customer:	 GEA Westfalia Separator Group GmbH	Benennung/Description: Separator	GEA IO 4			= _1_001			
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			Tested	19.06.2018/GR				Projekt: Project:						
Changes	Date	Name			© This document is COPYRIGHTED by GEA Westfalia Separator Group GmbH. All rights being reserved.			Sprache: Language:		en	Zeichnungs Nr.: Drawing No.: 8552-7300-086		DCC & EFS	Bl./Sh. 21 104 Bl./Sh.



			Date	19.06.2018	Kunde: - Customer:	 GEA Westfalia Separator Group GmbH	Benennung/Description: Lubrication oil pump -X11	GEA IO 4			= _3_002			
			Editor	HAN/PLE				Projekt: Project:			+ MCC			
			Tested	19.06.2018/GR										
Changes	Date	Name			© This document is COPYRIGHTED by GEA Westfalia Separator Group GmbH. All rights being reserved.			Sprache: Language:		en	Zeichnungs Nr.: Drawing No.:	8552-7300-086	DCC & EFS	Bl./Sh. 22 104 Bl./Sh.



14 12 11 /23.1

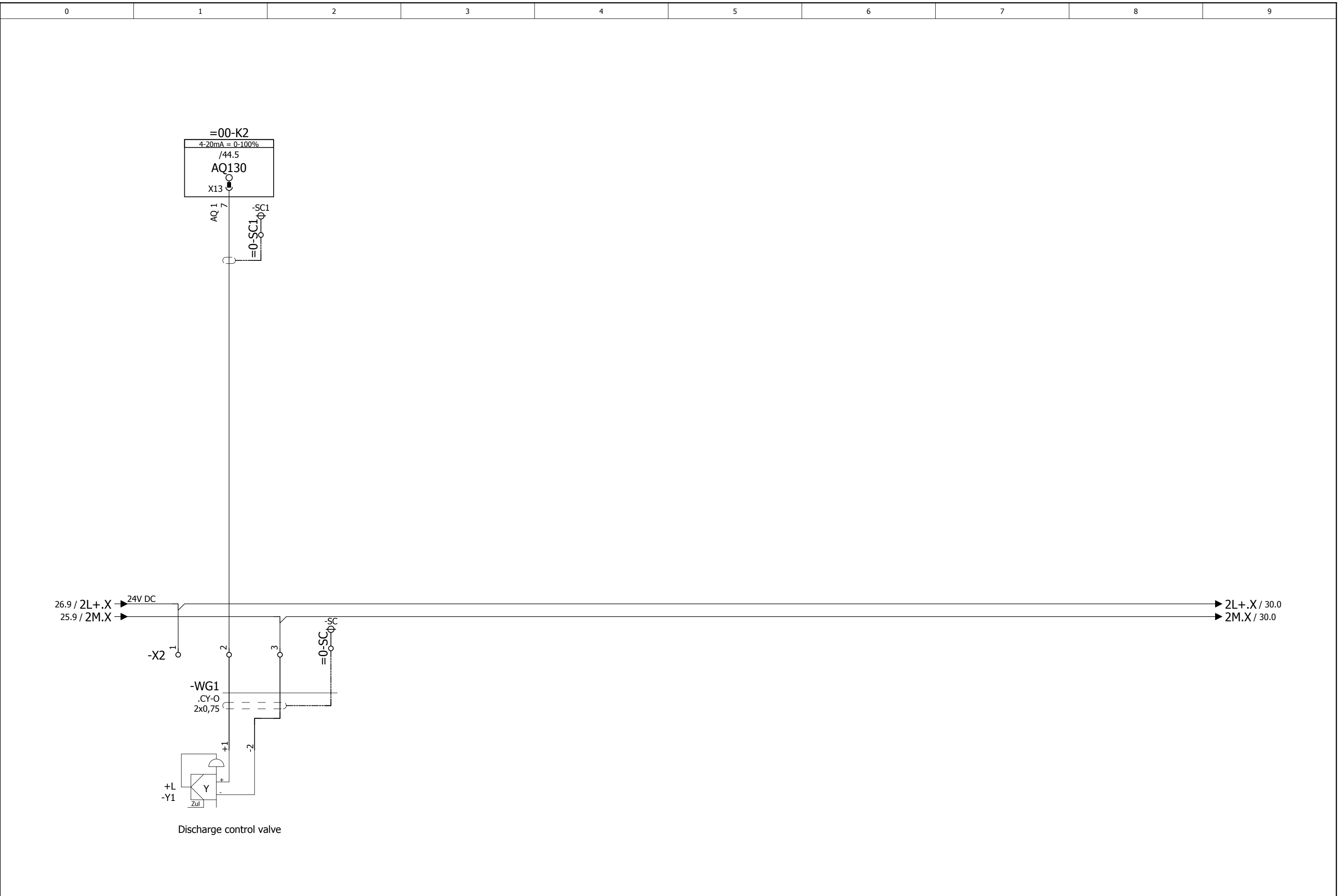
Sight glass illumination on / off

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			Editor	HAN/PLE						+ MCC		
			Tested	19.06.2018/GR				Sprache: Language:		Zeichnungs Nr.: Drawing No.:		DCC &EFS
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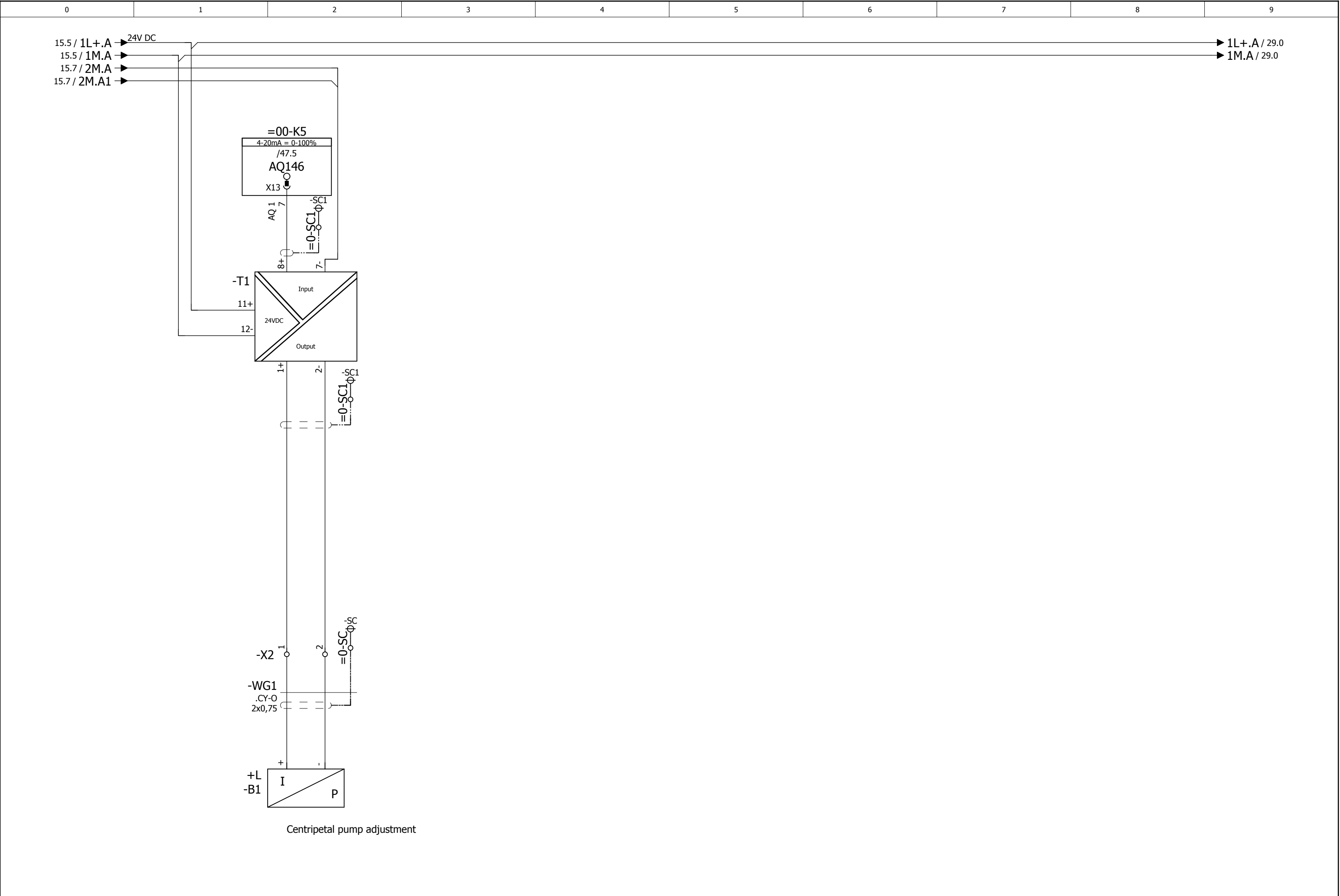


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			Tested	19.06.2018/GR	Projekt: Project:			Sprache: Language:		Zeichnungs Nr.: Drawing No.:		DCC &EFS	Bl./Sh. 24 104 Bl./Sh.
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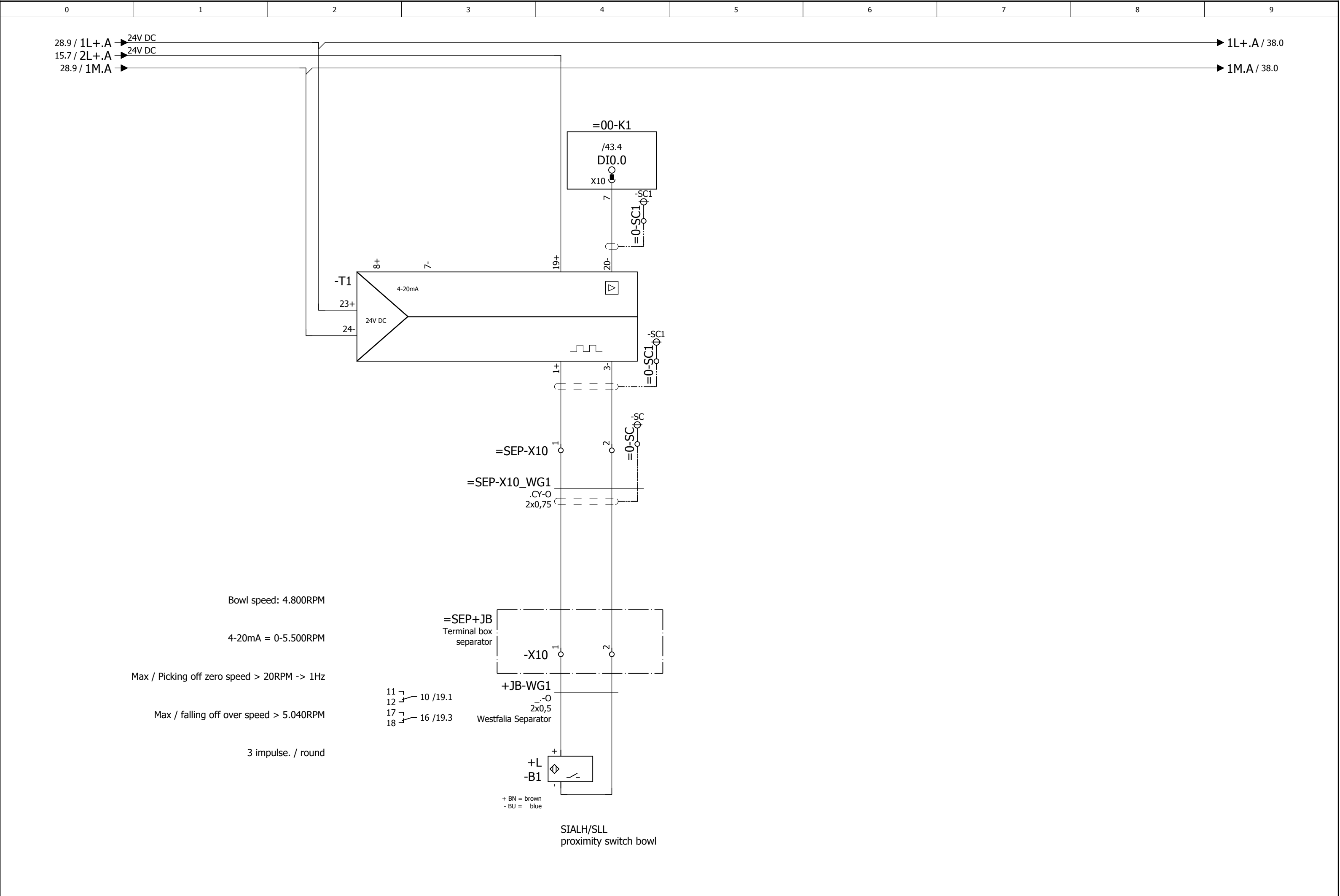


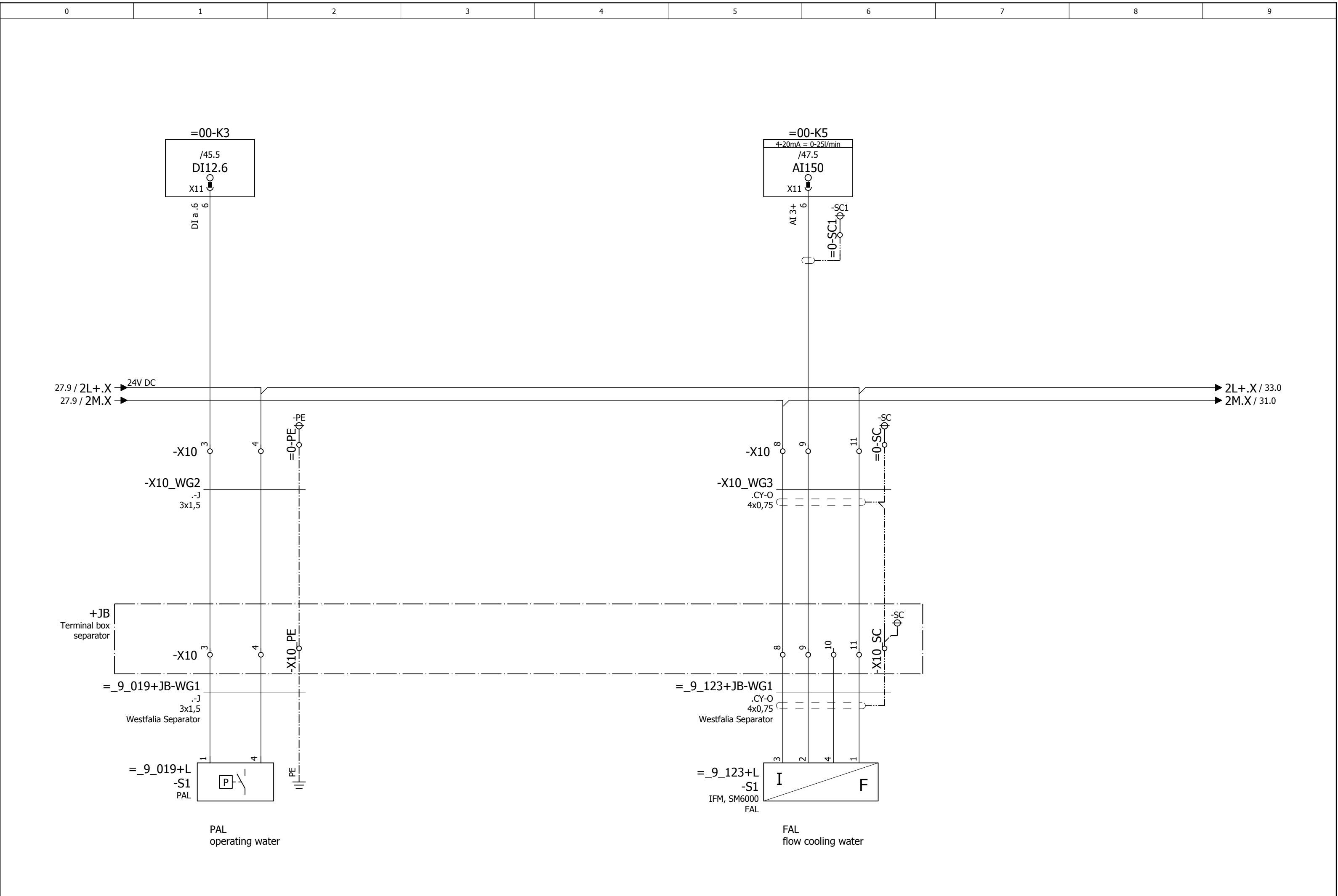


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			Tested	19.06.2018/GR	Projekt:			Sprache:		Zeichnungs Nr.:		DCC	Bl./Sh.	27
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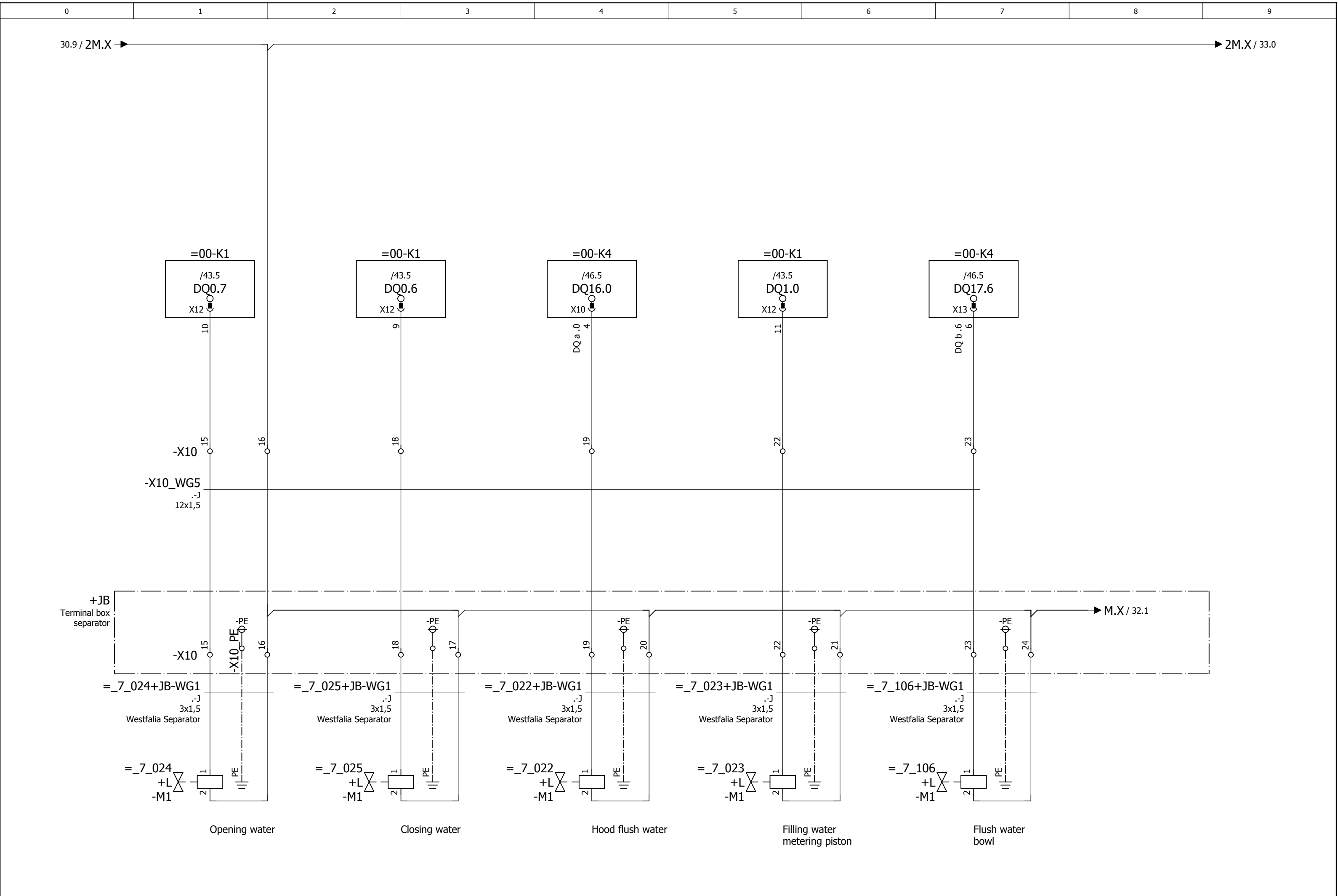


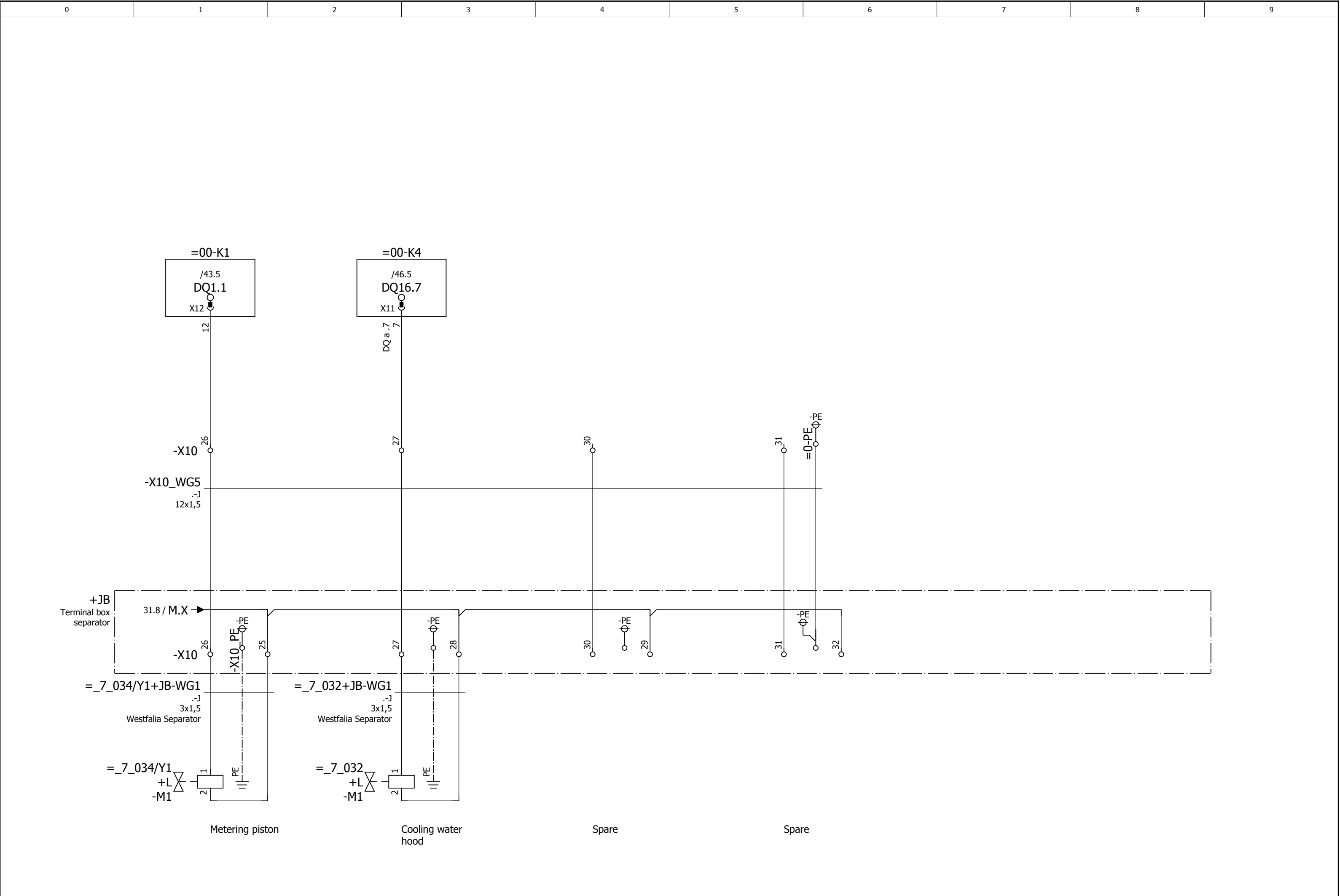
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			Tested	19.06.2018/GR	Projekt: Project:				Sprache: Language:		Zeichnungs Nr.: Drawing No.: 8552-7300-086		DCC	Bl./Sh. 28
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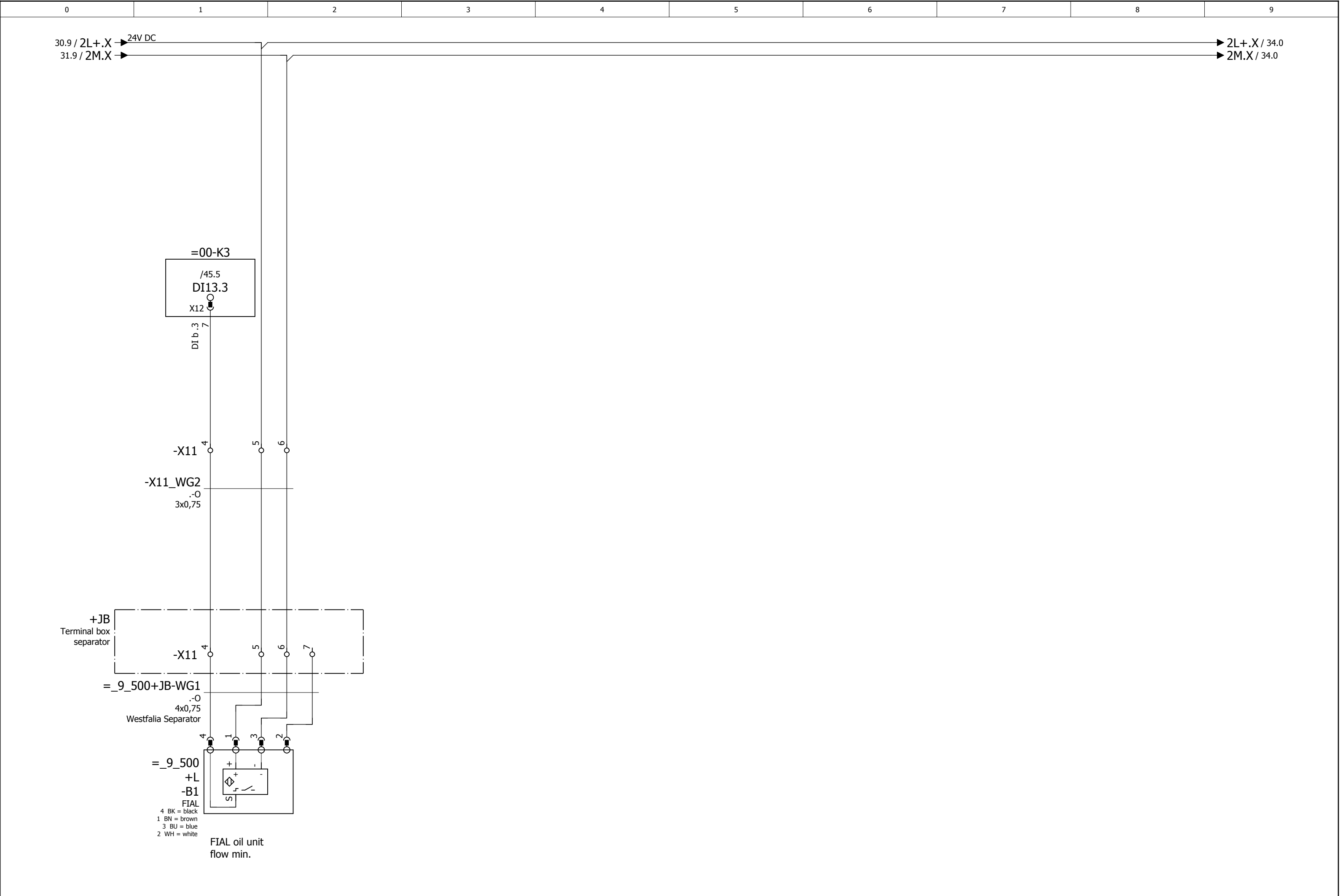




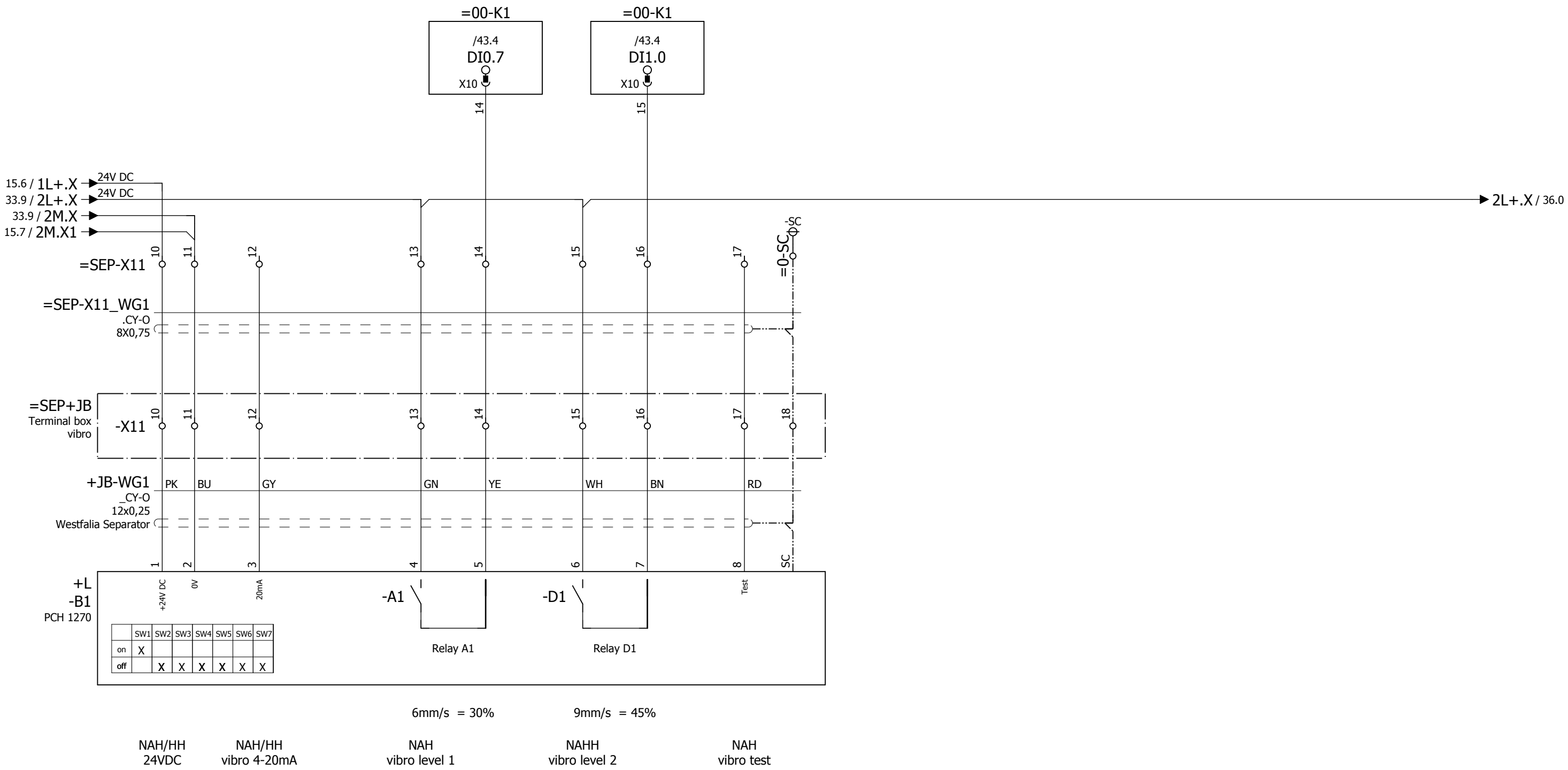
			Date	19.06.2018	Kunde: -	 GEA Westfalia Separator Group GmbH	Benennung/Description: Messages separator	GEA IO 4			= SEP		
			Editor	HAN/PLE	Customer:					+ PLC			
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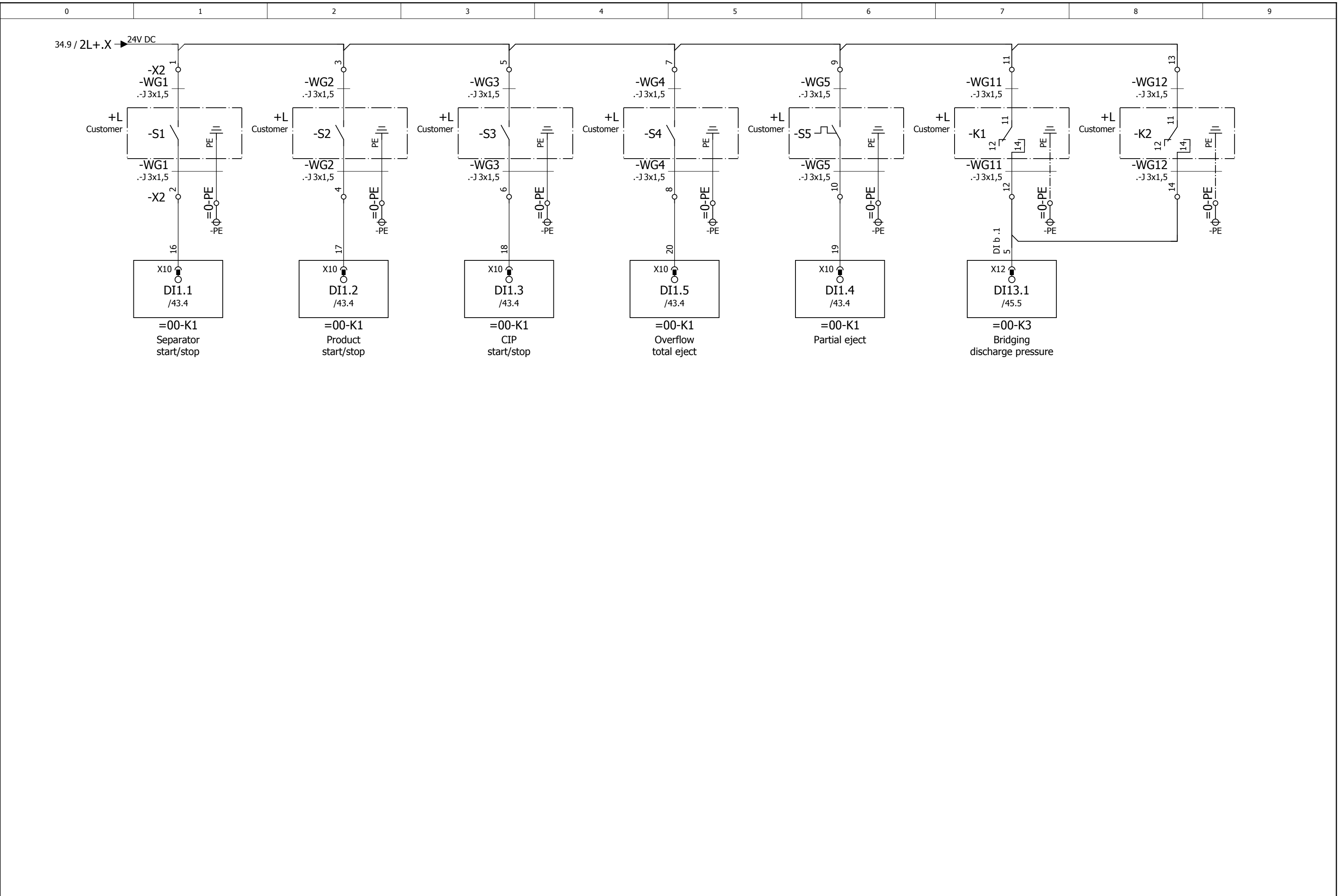


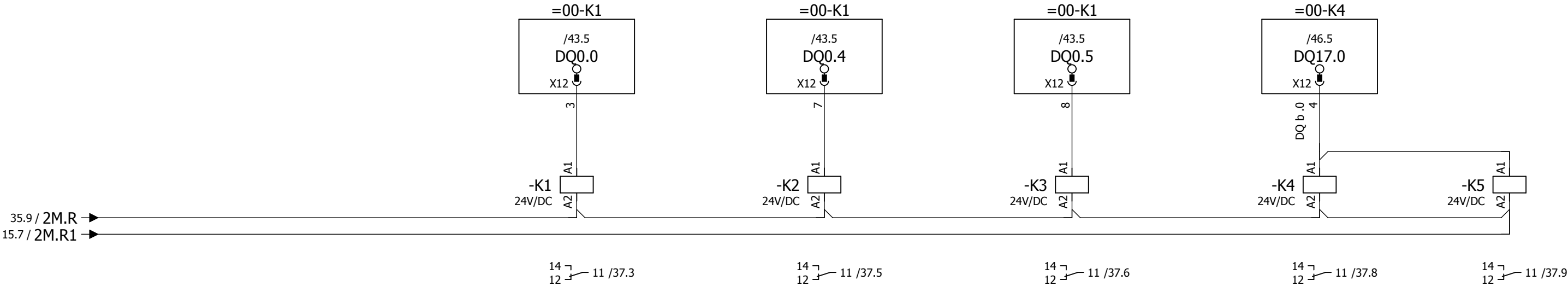




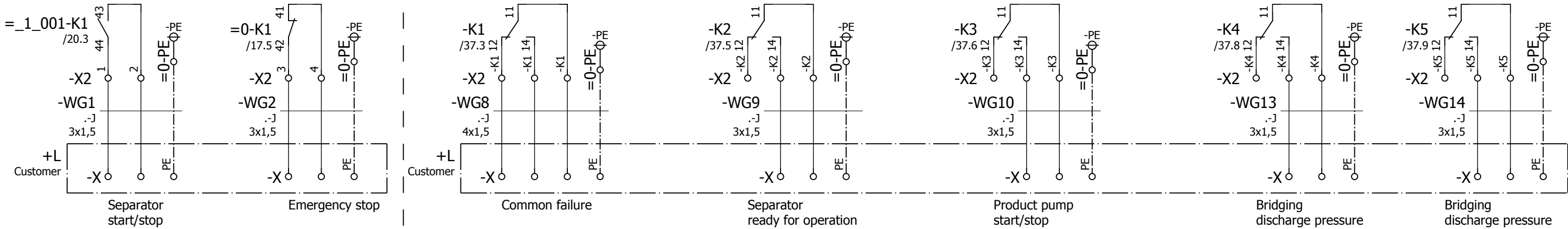
			Date	19.06.2018	Kunde: -	 GEA Westfalia Separator Group GmbH	Benennung/Description: Messages oil unit	GEA IO 4			= SEP		
			Editor	HAN/PLE	Customer:					+ PLC			
			Tested	19.06.2018/GR	Projekt: Project:			Sprache: Language:		Zeichnungs Nr.: Drawing No.:		DCC &EFS	Bl./Sh. 33 104 Bl./Sh.
Changes	Date	Name			© This document is COPYRIGHTED by GEA Westfalia Separator Group GmbH. All rights being reserved.			en	8552-7300-086				

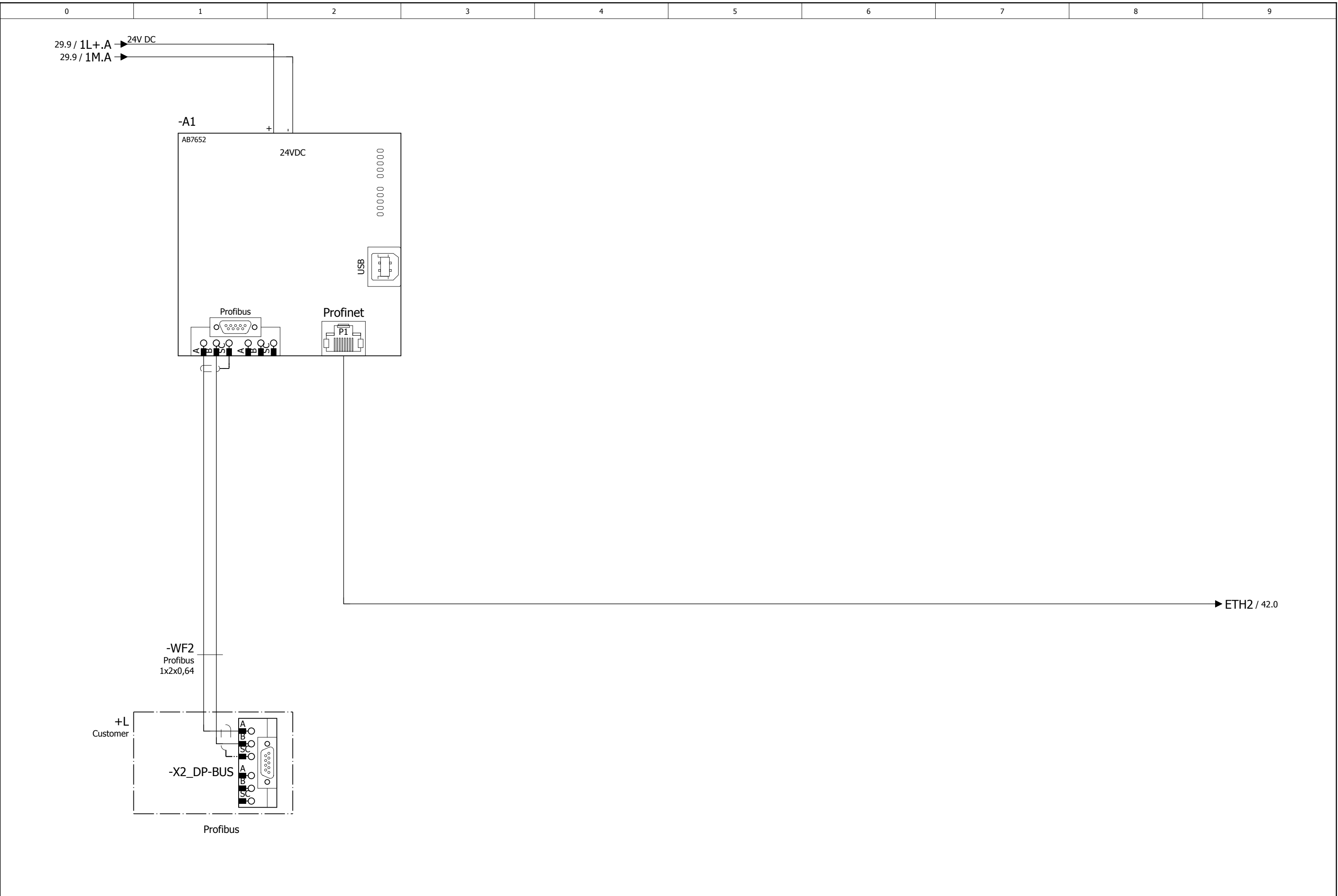




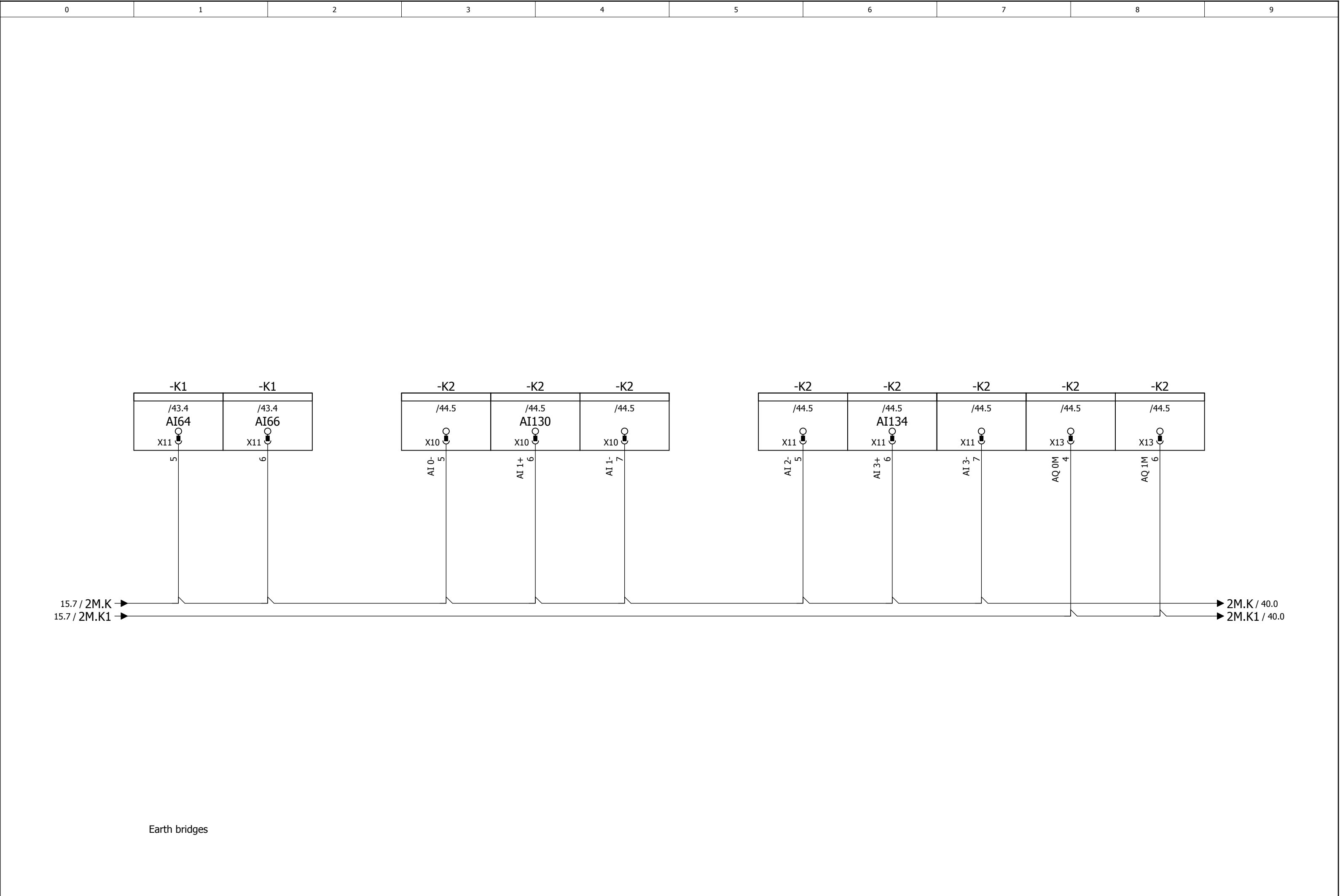


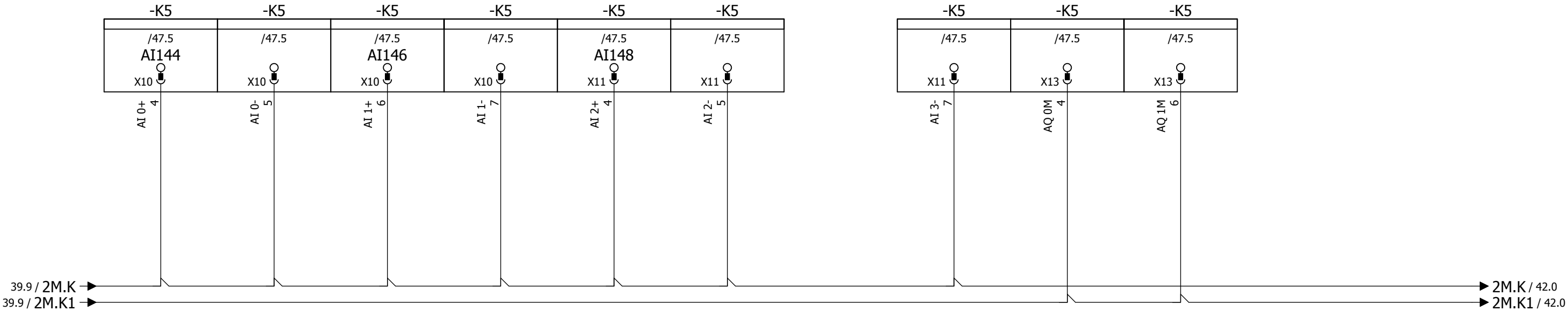
+MCC





			Date	19.06.2018	Kunde: -	 GEA Westfalia Separator Group GmbH	Benennung/Description: Gateway	GEA IO 4			= 00		
			Editor	HAN/PLE	Customer:						+ PLC		
			Tested	19.06.2018/GR	Projekt: Project:								
Changes	Date	Name			© This document is COPYRIGHTED by GEA Westfalia Separator Group GmbH. All rights being reserved.			Sprache: Language:		en	Zeichnungs Nr.: Drawing No.:	8552-7300-086	DCC & EFS

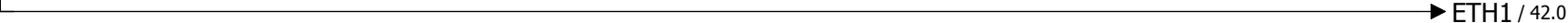
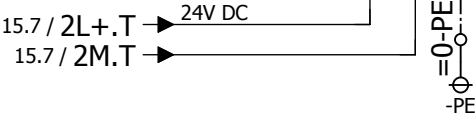
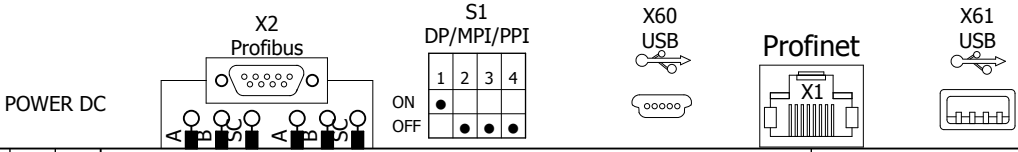




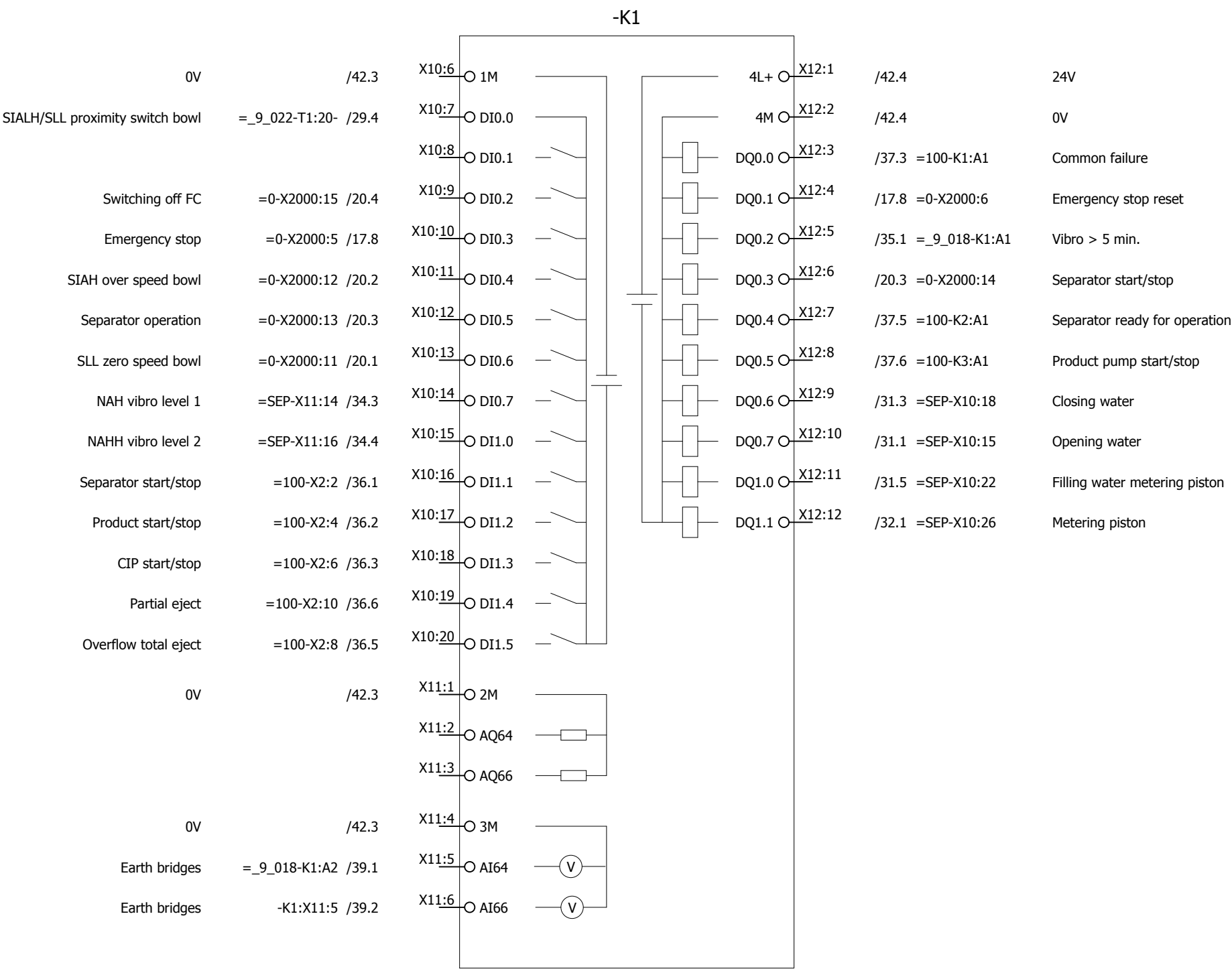
Earth bridges

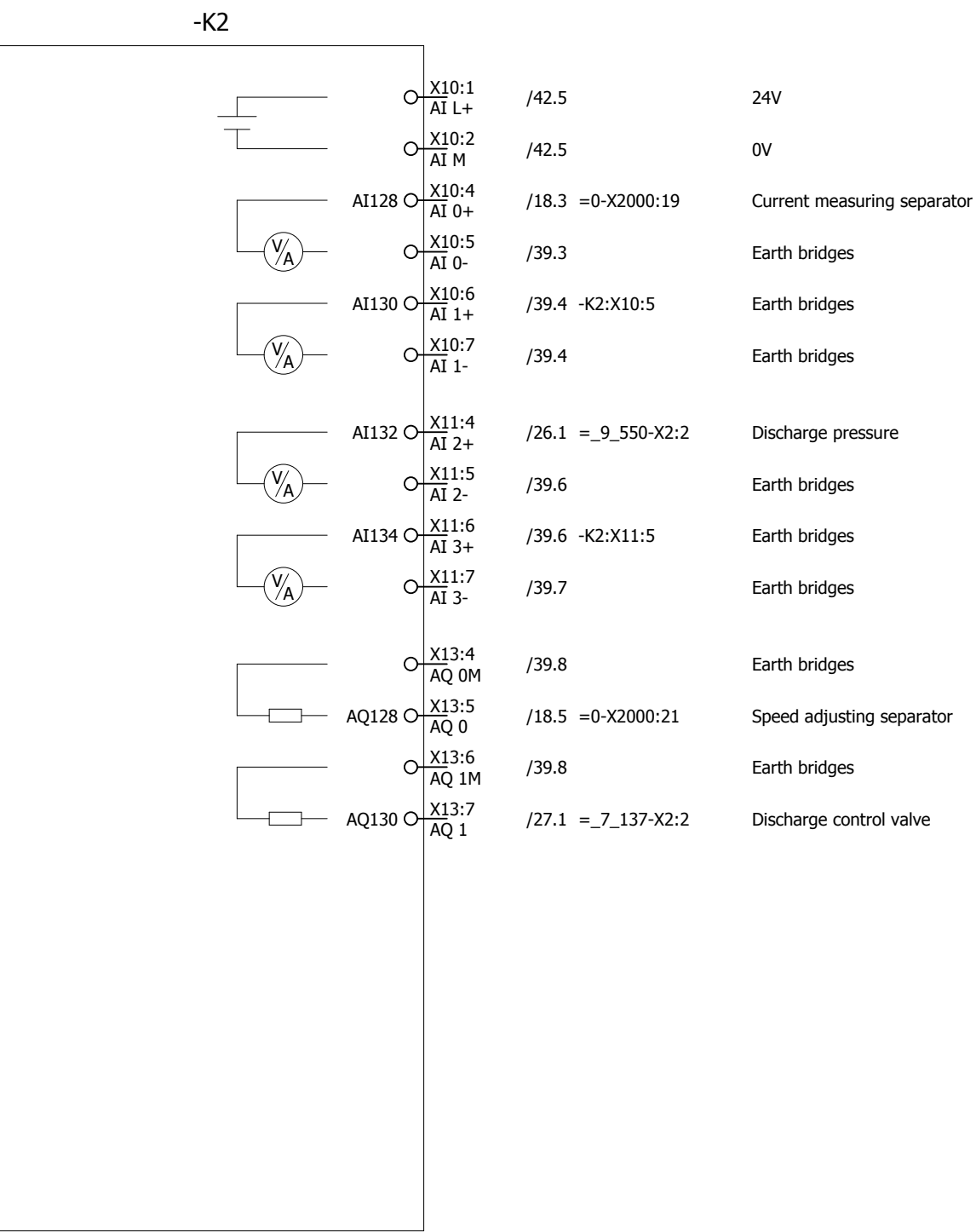
-P1

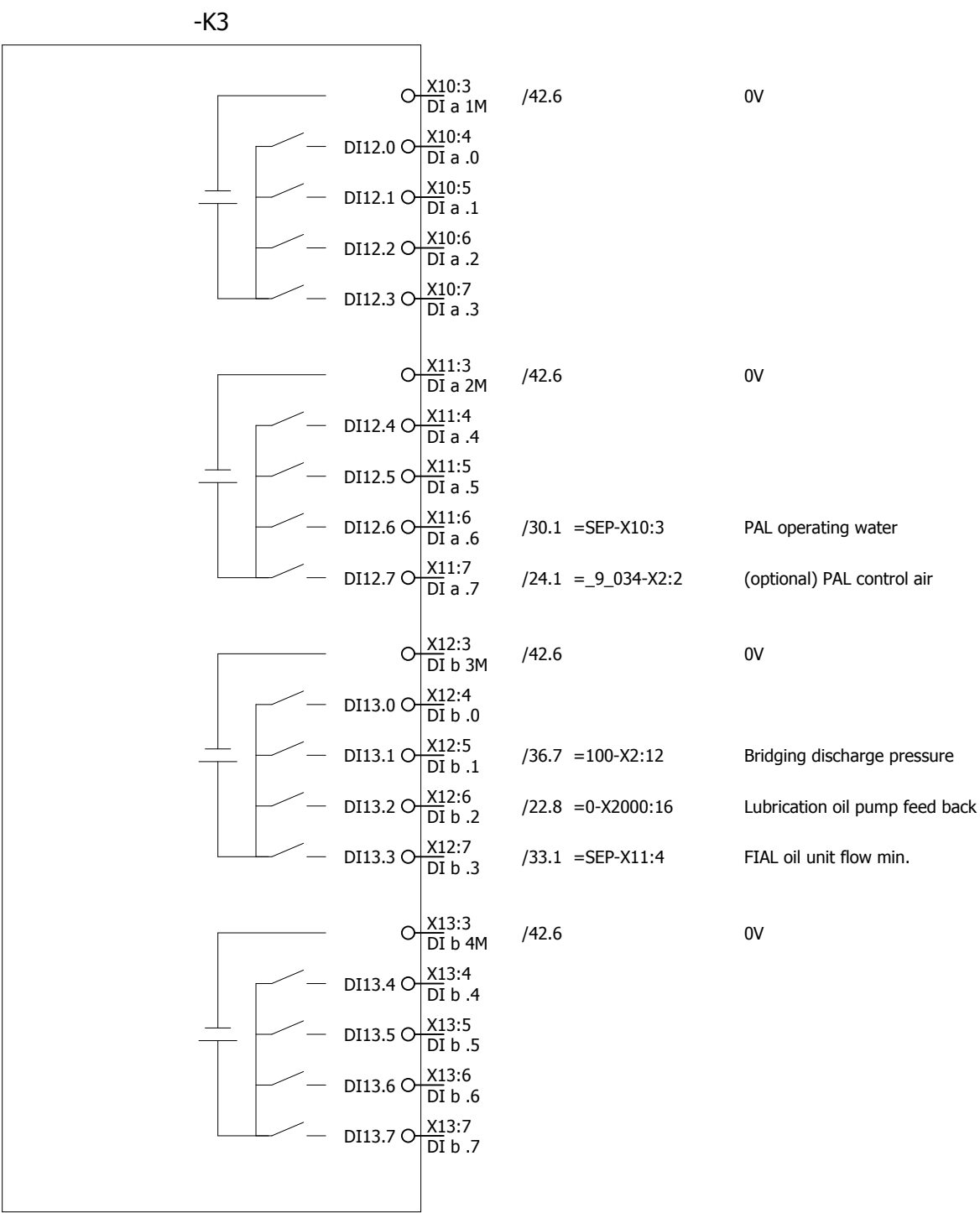
GEA IO 4

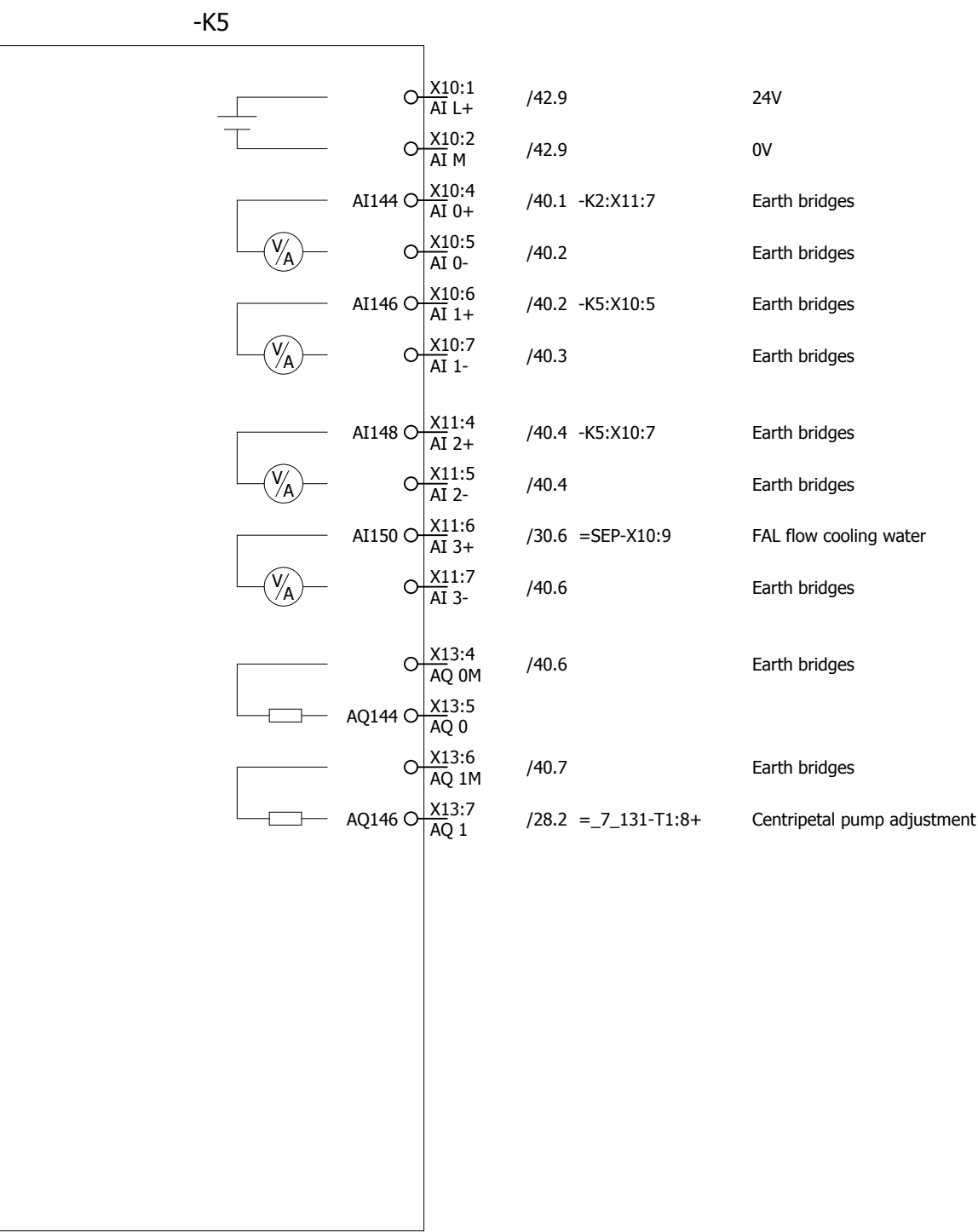


Operating panel KTP400

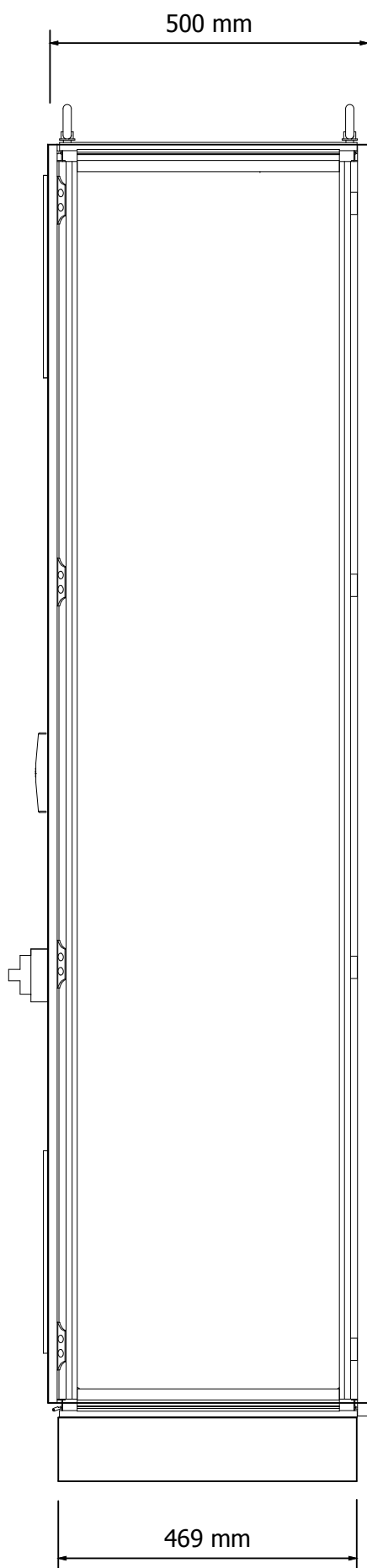
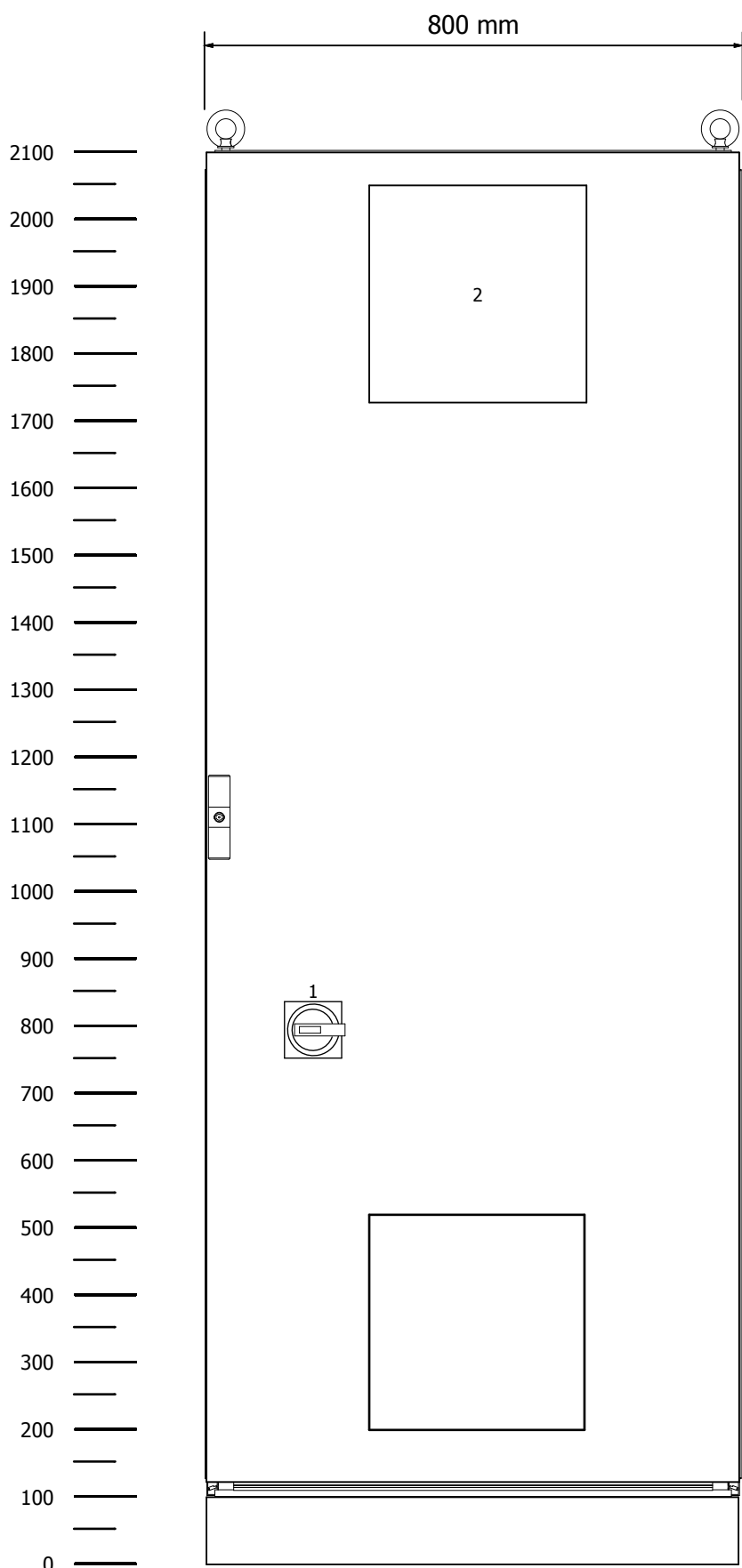






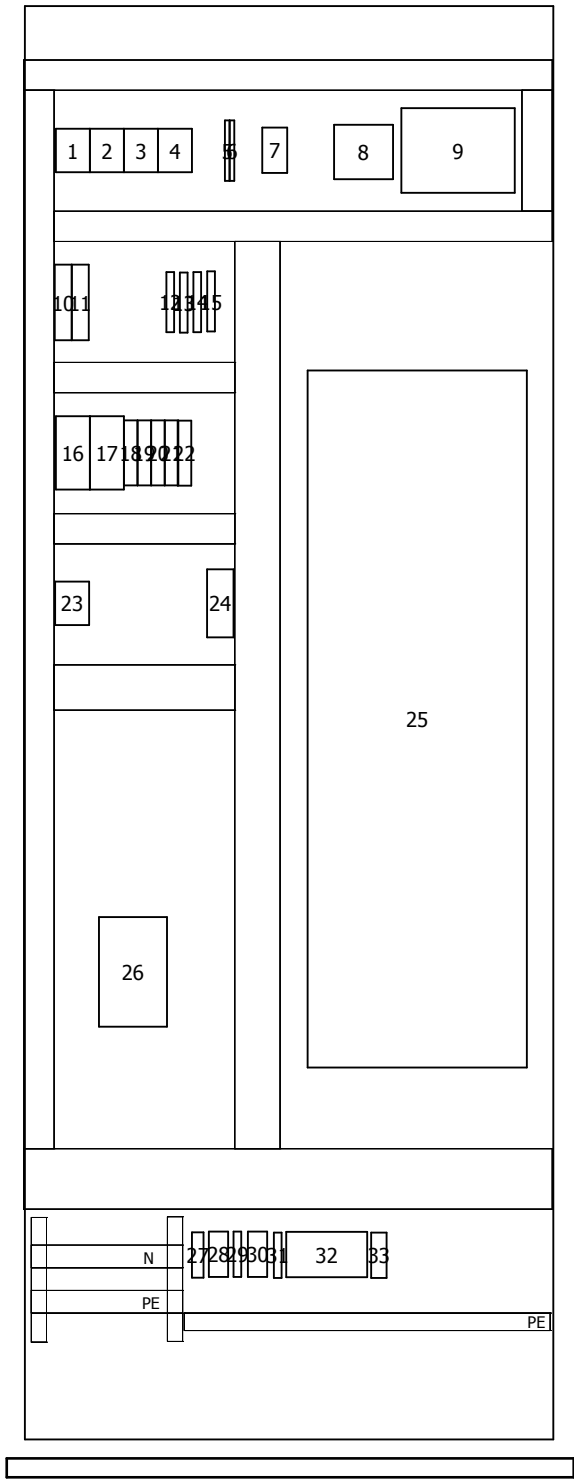


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1 =0-Q1 Isolator
2 =0-M1 Cabinet fan

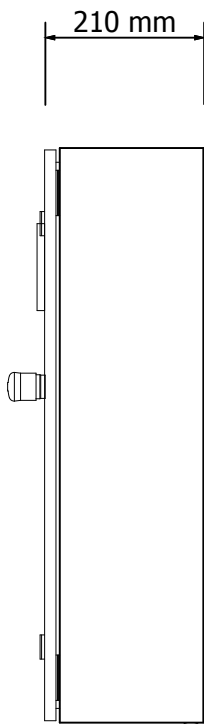
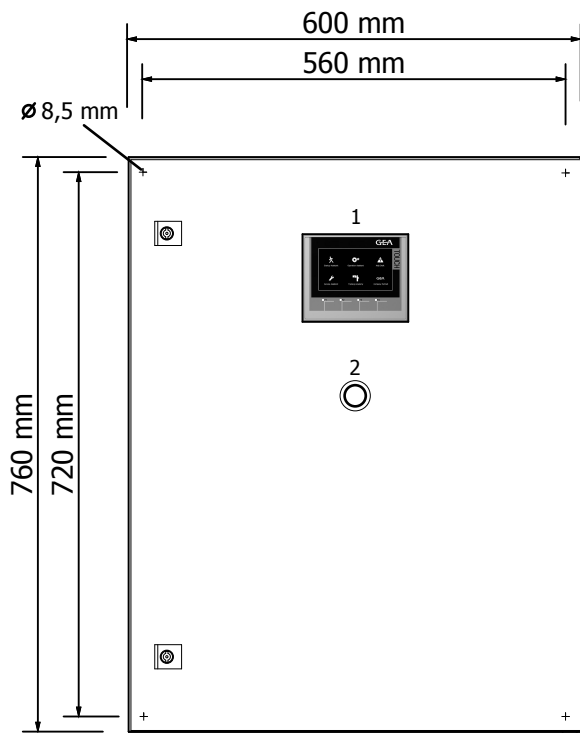
			Date	19.06.2018	Kunde: - Customer: Projekt: Project:	 GEA Westfalia Separator Group GmbH	Benennung/Description: Arrangement drawing	GEA IO 4			=			
			Editor	HAN/PLE						+ MCC				
			Tested	19.06.2018/GR										
Changes	Date	Name			© This document is COPYRIGHTED by GEA Westfalia Separator Group GmbH. All rights being reserved.			Sprache: Language:		en	Zeichnungs Nr.: Drawing No.:	8552-7300-086	DCC & ELU	Bl./Sh. 48 104 Bl./Sh.



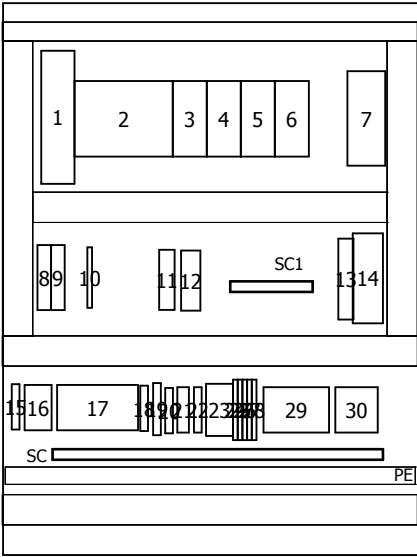
- 1 =0-K3
- 2 =_1_001-K1
- 3 =_1_001-K3
- 4 =_1_001-K4
- 5 =_3_002-K1
- 6 =_8-K1
- 7 =0-B1
- 8 =_8-T1
- 9 =0-T1
- 10 =0-K1
- 11 =_1_001-K2
- 12 =0-X1.1
- 13 =0-X1.2
- 14 =0-X2.1
- 15 =0-X2.2
- 16 =_3_002-F1
- 17 =0-F1
- 18 =0-F2
- 19 =0-F4
- 20 =0-F5
- 21 =0-F6
- 22 =_8-F1
- 23 =_3_002-Q1
- 24 =_1_001-P1
- 25 =_1_001-T1
- 26 =0-Q1
- 27 =SEP-X11.1
- 28 =0-X1
- 29 =0-X1000
- 30 =_8-X1
- 31 =0-X2
- 32 =0-X2000
- 33 =100-X2

Isolator

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- 1 =00-P1
- 2 =0-S0
- Operating panel KTP400
- Emergency stop



- 1 =00-A1
- 2 =00-K1
- 3 =00-K2
- 4 =00-K3
- 5 =00-K4
- 6 =00-K5
- 7 =0-T2
- 8 =0-F2
- 9 =0-F7
- 10 =_9_018-K1
- 11 =0-X2.1
- 12 =0-X2.2
- 13 =_7_131-T1
- 14 =_9_022-T1
- 15 =0-X1000
- 16 =0-X2
- 17 =0-X2000
- 18 =_9_034-X2
- 19 =Y-X2
- 20 =_9_550-X2
- 21 =_7_137-X2
- 22 =_7_131-X2
- 23 =100-X2
- 24 =100-K1
- 25 =100-K2
- 26 =100-K3
- 27 =100-K4
- 28 =100-K5
- 29 =SEP-X10
- 30 =SEP-X11

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																Cable type	Terminal strip designation =0+MCC-N																Cable type												
Function text																Cable name	Target designation				Connection	Terminal	Link	Target designation				Connection	Cable name									Page / column							
Incoming																		-PE						PE	N	Ⓝ	-N														&EFS/12.2 SV9340.040				
				Date		19.06.2018		Kunde: -				 GEA Westfalia Separator Group GmbH				Benennung/Description: Terminal connection diagram =0+MCC-N								GEA IO 4				=																	
				Editor		HAN/PLE		Customer:																								+ MCC													
				Tested		19.06.2018/GR		Projekt:																																					
Changes				Date		Name		© This document is COPYRIGHTED by GEA Westfalia Separator Group GmbH. All rights being reserved.												Sprache: Language: en				Zeichnungs Nr.: Drawing No.: 8552-7300-086				DCC &EMA		Bl./Sh. 52 104 Bl./Sh.															

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												Terminal strip designation																
												=0+MCC-PE																
Function text						=100+MCC WG2	=100+MCC WG1	=_8_006+MCC WG1	=_8_005+MCC WG1	=SEP+MCC X11	=0+PLC WG2000	=0+PLC WD1000	=0+MCC WD1															
													PE															

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																Cable type	Terminal strip designation =0+MCC-X1.1																Cable type														
Function text																	Cable name	Target designation				Connection	Terminal	Link	Target designation				Connection	Cable name					Page / column												
Control voltage AC air conditioning																										1		-F4				2						&EFS/14.1 ZDU 2,5/4AN									
=																						1	-B1					1						&EFS/14.1													
Control voltage AC																						2	-F5					2						&EFS/14.2 ZDU 2,5/4AN													
=																						2	=_1_001-K1					23						&EFS/14.2													
				Date				19.06.2018				Kunde: -				 GEA Westfalia Separator Group GmbH				Benennung/Description: Terminal connection diagram =0+MCC-X1.1								GEA IO 4								=											
				Editor				HAN/PLE				Customer:																												DCC				Bl./Sh. 55			
				Tested				19.06.2018/GR				Project:																																			
Changes				Date				Name								© This document is COPYRIGHTED by GEA Westfalia Separator Group GmbH. All rights being reserved.																															

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																	Terminal strip designation																																																								
																	=0+MCC-X1.2																																																								
Function text																		Target designation																				Target designation																				Page / column															
Control voltage AC air conditioning																																						-X1000																				&EFS/14.1															
=																																						-M1																				ZDU 2,5/4AN &EFS/14.1															
Control voltage AC																		=_3_002-Q1																	A2																							ZDU 2,5/4AN &EFS/14.2															
=																																						=_1_001-P1																				&EFS/14.2															
																</																																																									

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																	Terminal strip designation																																
																	=0+MCC-X2																																
Function text																	Target designation																	Page / column															
Emergency stop																	1 ' -X2000 4																	&EFS/17.3															
=																	2 ' -K1 A1																	&EFS/17.3															

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																Cable type	Terminal strip designation =0+MCC-X2.1																Cable type												
Function text																	Cable name	Target designation				Connection	Terminal	Link	Target designation				Connection	Cable name									Page / column						
Control voltage DC control																		-X2000							3	1		-F6				2						&EFS/15.1 ZDU 2,5/4AN							
=																					-K3							33						&EFS/15.1											
Control voltage DC PLC																	=_3_002-Q1				13	2												&EFS/15.2 ZDU 2,5/4AN											
=																								=_1_001-K3										33		&EFS/15.2									
				Date		19.06.2018		Kunde: -				 GEA Westfalia Separator Group GmbH				Benennung/Description: Terminal connection diagram =0+MCC-X2.1								GEA IO 4				=																	
				Editor		HAN/PLE		Customer:																								+ MCC													
				Tested		19.06.2018/GR		Projekt:																																					
Changes				Date		Name						© This document is COPYRIGHTED by GEA Westfalia Separator Group GmbH. All rights being reserved.												Sprache: Language: en				Zeichnungs Nr.: Drawing No.: 8552-7300-086				DCC &EMA		Bl./Sh. 58 104 Bl./Sh.											

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																Cable type	Terminal strip designation =0+MCC-X2.2																Cable type																
Function text																	Cable name	Target designation				Connection	Terminal	Link	Target designation				Connection	Cable name																			
Control voltage DC control																									A2	1		-X2000				2																	
=																					-K1							A2																					
Control voltage DC PLC																	=_3_002-K1																																
=																					-K3								A2																				

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																			Terminal strip designation =0+MCC-X2000																								
Function text																																	Page / column										
Control voltage DC																																	&EFS/13.7 WDU 2,5										
=																																	&EFS/13.7 WDU 2,5										
Emergency stop																																	&EFS/17.1 WDU 2,5										
=																																	&EFS/17.1 WDU 2,5										
=																																	&EFS/17.8 WDU 2,5										
Emergency stop reset																																	&EFS/17.8 WDU 2,5										
Zero speed bowl																																	&EFS/19.1 WDU 2,5										
=																																	&EFS/19.1 WDU 2,5										
Over speed bowl																																	&EFS/19.3 WDU 2,5										
=																																	&EFS/19.3 WDU 2,5										
SLL zero speed bowl																																	&EFS/20.1 WDU 2,5										
SIAH over speed bowl																																	&EFS/20.2 WDU 2,5										
Separator operation																																	&EFS/20.3 WDU 2,5										
Separator start/stop																																	&EFS/20.3 WDU 2,5										
Switching off FC																																	&EFS/20.4 WDU 2,5										
Lubrication oil pump feed back																																	&EFS/22.8 WDU 2,5										
Lubrication oil pump start/stop																																	&EFS/22.8 WDU 2,5										
Sight glass illumination on / off																																	&EFS/23.8 WDU 2,5										
Current measuring separator																																	&EFS/18.3 WDU 2,5										
=																																	&EFS/18.4 WDU 2,5										
Speed adjusting separator																																	&EFS/18.5 WDU 2,5										
				Date		19.06.2018		Kunde: –								GEA Westfalia				Benennung/Description:								GEA IO 4				=											
				Editor		HAN/PLE		Customer:								Separator Group GmbH				Terminal connection diagram =0+MCC-X2000												+ MCC											
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Changes				Date		Name		© This document is COPYRIGHTED by GEA Westfalia Separator Group GmbH. All rights being reserved.																				Language: en				Drawing No.: 8552-7300-086				&EMA				104 Bl./Sh.			

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Function text								=100+PLC WG11	=100+PLC WG5	=100+PLC WG4	=100+PLC WG3	=100+PLC WG2	=100+PLC WG1	=SEP+PLC X10_WG5	=SEP+PLC X10_WG2	=0+PLC WG2000	=0+PLC WD1000	Cable name	Cable type	Terminal strip designation =0+PLC-PE								Cable type												
																								Target designation	Connection	Terminal	Link	Target designation	Connection											Page / column
Power supply																				-PE		PE	⊕	-T2	PE										&EFS/13.5					
Control voltage AC																	PE			+MCC-PE	PE	PE	⊕	-PE											&EFS/13.6					
Control voltage DC																				-PE		PE	⊕	-X2	5										&EFS/13.6					
PE rail																				-SC	SC	PE	⊕	-PE											&EFS/13.7					
Control voltage DC																	PE			+MCC-PE	PE	PE	⊕	-PE											&EFS/13.7					
PE rail																				-SC1	SC1	PE	⊕	-PE											&EFS/13.7					
Earthing mounting plate 1																			-1M	PE	PE	⊕	-PE												&EFS/13.8					
Earthing door 1																			-1T	PE	PE	⊕	-PE												&EFS/13.8					
Earthing cabinet 1																			-1G	PE	PE	⊕	-PE												&EFS/13.9					
Emergency stop																					PE	⊕	-PE													&EFS/17.1				
(optional) PAL control air																						PE	⊕	-PE												&EFS/24.2				
(optional) product feed																						PE	⊕	-PE												&EFS/25.1				
(optional) water feed																						PE	⊕	-PE												&EFS/25.2				
PAL operating water																PE			=SEP+JB-X10_PE	PE	PE	⊕	-PE												&EFS/30.2					
Spare														PE					=SEP+JB-X10_PE	PE	PE	⊕	-PE												&EFS/32.6					
Separator start/stop													PE						=100+L-S1	PE	PE	⊕	-PE												&EFS/36.1					
Product start/stop												PE							=100+L-S2	PE	PE	⊕	-PE												&EFS/36.2					
CIP start/stop											PE								=100+L-S3	PE	PE	⊕	-PE												&EFS/36.4					
Overflow total eject										PE									=100+L-S4	PE	PE	⊕	-PE												&EFS/36.5					
Partial eject									PE										=100+L-S5	PE	PE	⊕	-PE												&EFS/36.6					
Bridging discharge pressure								PE											=100+L-K1	PE	PE	⊕	-PE												&EFS/36.7					

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										Terminal strip designation											
										=0+PLC-PE											
Function text																					
Bridging discharge pressure																					
Common failure																					
Separator ready for operation																					
Product pump start/stop																					
Bridging discharge pressure																					
=																					
Operating panel KTP400																					
CPU																					
Analog input card / output card																					
Digital input card																					
Digital output card																					
Analog input card / output card																					

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																Cable type	Terminal strip designation =0+PLC-SC1																Cable type										
Function text																	Cable name	Target designation				Connection	Terminal	Link	Target designation				Connection	Cable name					Page / column								
SC rail																		-PE				PE	SC1	┆	-SC1											&EFS/13.8							
Current measuring separator																						SC1	┆	-SC1									&EFS/18.3										
Speed adjusting separator																						SC1	┆	-SC1									&EFS/18.5										
Discharge pressure																						SC1	┆	-SC1									&EFS/26.1										
Discharge control valve																						SC1	┆	-SC1									&EFS/27.1										
Centripetal pump adjustment																						SC1	┆	-SC1										&EFS/28.2									
=																						SC1	┆	-SC1										&EFS/28.2									
SIALH/SLL proximity switch bowl																						SC1	┆	-SC1										&EFS/29.4									
=																						SC1	┆	-SC1										&EFS/29.4									
FAL flow cooling water																						SC1	┆	-SC1										&EFS/30.6									
				Date		19.06.2018		Kunde: -				 GEA Westfalia Separator Group GmbH				Benennung/Description: Terminal connection diagram =0+PLC-SC1								GEA IO 4				=															
				Editor		HAN/PLE		Customer:																								+ PLC											
				Tested		19.06.2018/GR		Projekt:																																			
Changes				Date		Name						© This document is COPYRIGHTED by GEA Westfalia Separator Group GmbH. All rights being reserved.												Sprache: Language:				en				Zeichnungs Nr.: Drawing No.: 8552-7300-086				DCC &EMA				Bl./Sh. 70 104 Bl./Sh.			

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																Cable type	Terminal strip designation =0+PLC-X2.1																Cable type														
Function text																	Cable name	Target designation				Connection	Terminal	Link	Target designation				Connection	Cable name					Page / column												
Control voltage DC control																					1	●	-F7	2										&EFS/15.5 ZDU 2,5/4AN													
=																				1	●	=_7_131-T1	11+								&EFS/15.5																
=																		=SEP-X11	10	2	●									&EFS/15.6 ZDU 2,5/4AN																	
=																				2	●									&EFS/15.6																	
Control voltage DC PLC																				3	●									&EFS/15.6 ZDU 2,5/4AN																	
=																	=00-P1	X80:L+	3	●	=_9_022-T1	19+									&EFS/15.7																
=																	=_9_034-X2	1	4	●								&EFS/15.7 ZDU 2,5/4AN																			
=																			4	●	=00-K1	X10:1						&EFS/15.7																			
				Date				19.06.2018				Kunde: -				GEA Westfalia Separator Group GmbH				Benennung/Description: Terminal connection diagram =0+PLC-X2.1								GEA IO 4				=															
				Editor				HAN/PLE				Customer:																								+ PLC											
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Control voltage DC control																										1		-X2000				2						&EFS/15.5 ZDU 2,5/4AN			
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=																	=_9_018-K1				A2	4		=00-K1				X11:5						&EFS/15.7							
=																	=SEP-X11				11	5		=_7_131-T1				7-						&EFS/15.7 ZDU 2,5/4AN							
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Changes				Date		Name						© This document is COPYRIGHTED by GEA Westfalia Separator Group GmbH. All rights being reserved.								Sprache:				Zeichnungs Nr.:				DCC				Bl./Sh.									
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Function text														=SEP+PLC X10_WG5	=SEP+PLC X10_WG3	=SEP+PLC X10_WG2	=SEP+PLC X10_WG1	Cable type		Target designation	Connection	Terminal	Link	Target designation	Connection	Cable name							Page / column						
SIALH/SLL proximity switch bowl																1		+JB-X10	1	1		=_9_022-T1	1+								&EFS/29.4								
=																2		+JB-X10	2	2		=_9_022-T1	3-								&EFS/29.4								
PAL operating water															1		+JB-X10	3	3		=00-K3	X11:6								&EFS/30.1									
=															2		+JB-X10	4	4	●	=_7_137-X2	1							&EFS/30.1										
FAL flow cooling water														1		+JB-X10	8	8	●	=_7_137-X2	3								&EFS/30.5										
=														2		+JB-X10	9	9		=00-K5	X11:6								&EFS/30.6										
=														3		+JB-X10	11	11	●	-X11	5								&EFS/30.6										
Opening water														1		+JB-X10	15	15		=00-K1	X12:10									&EFS/31.1									
=														2		+JB-X10	16	16	●	-X11	6								&EFS/31.2										
Closing water														3		+JB-X10	18	18		=00-K1	X12:9									&EFS/31.3									
Hood flush water														4		+JB-X10	19	19		=00-K4	X10:4									&EFS/31.4									
Filling water metering piston														5		+JB-X10	22	22		=00-K1	X12:11									&EFS/31.5									
Flush water bowl														6		+JB-X10	23	23		=00-K4	X13:6									&EFS/31.7									
Metering piston														7		+JB-X10	26	26		=00-K1	X12:12									&EFS/32.1									
Cooling water hood														8		+JB-X10	27	27		=00-K4	X11:7									&EFS/32.3									
Spare														9		+JB-X10	30	30												&EFS/32.4									
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																			=SEP+JB-X10																				
Target designation								Connection	Terminal	Link	Target designation				Connection	Cable name	=SEP+PLC X10_WG1	=SEP+PLC X10_WG2	=SEP+PLC X10_WG3	=SEP+PLC X10_WG5			Page / column																
SIALH/SLL proximity switch bowl																BN		= _9_022+L-B1	+	1	┆	+PLC-X10				1		1						&EFS/29.4					
=															BU	= _9_022+L-B1		-	2	┆	+PLC-X10				2	2							&EFS/29.4						
PAL operating water															1			= _9_019+L-S1	1	3	┆	+PLC-X10				3			1					&EFS/30.1					
=															2			= _9_019+L-S1	4	4	┆	+PLC-X10				4			2					&EFS/30.1					
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FAL flow cooling water															1			= _9_123+L-S1	3	8	┆	+PLC-X10				8					1			&EFS/30.5					
=														2		= _9_123+L-S1		2	9	┆	+PLC-X10				9					2				&EFS/30.6					
=														3		= _9_123+L-S1		4	10	┆														&EFS/30.6					
=														4		= _9_123+L-S1		1	11	┆	+PLC-X10				11					3				&EFS/30.6					
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Opening water														1			= _7_024+L-M1	1	15	┆	+PLC-X10				15						1			&EFS/31.1					
=													2		= _7_024+L-M1		2	16	●	+PLC-X10				16						2			&EFS/31.2						
Closing water													2		= _7_025+L-M1		2	17	●														&EFS/31.3						
=													1		= _7_025+L-M1		1	18	┆	+PLC-X10				18						3			&EFS/31.3						
Hood flush water													1		= _7_022+L-M1		1	19	┆	+PLC-X10				19						4			&EFS/31.4						
=													2		= _7_022+L-M1	2	20	●													&EFS/31.4								
Filling water metering piston													2			= _7_023+L-M1	2	21	●													&EFS/31.6							
=													1			= _7_023+L-M1	1	22	┆	+PLC-X10				22					5			&EFS/31.5							
Flush water bowl													1			= _7_106+L-M1	1	23	┆	+PLC-X10				23					6			&EFS/31.7							
=													2			= _7_106+L-M1	2	24	●													&EFS/31.7							
Metering piston													2			= _7_034/Y1+L-M1	2	25	●													&EFS/32.2							
			Date	19.06.2018	Kunde: -	GEA Westfalia Separator Group GmbH				Benennung/Description: Terminal connection diagram =SEP+JB-X10										GEA IO 4						= + JB		DCC Bl./Sh. 85											
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Changes	Date	Name			© This document is COPYRIGHTED by GEA Westfalia Separator Group GmbH. All rights being reserved.														Sprache: en		Zeichnungs Nr.: 8552-7300-086				&EMA		104 Bl./Sh.												

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Cable layout

Cable name	/page.path	Source (from)	Target (to)	Cable type	Wires total	Wires used	Cross-section	Length	External diameter	Remark
=0+MCC-WD1	&EFS/12.1	=0+MCC-PE;=0+MCC-XQ1	=CUSTOMER+L-X	_-J	4x	4	70 mm²		35,0 mm	
=_1_001+MCC-WD1	&EFS/18.1	=_1_001+MCC-XT1	=_1_001+L-M1	FC.CY	3x	3	50+50 mm²		30,0 mm	
=_1_001+MCC-WG1	&EFS/18.3	=_1_001+MCC-XT1	=_1_001+L-M1	.CY-O	2x	2	0,75 mm²		7,4 mm	
=_8_005+MCC-WG1	&EFS/23.1	=0+MCC-PE;=_8+MCC-X1	=_8_005+L-H1	.-J	3x	3	1,5 mm²		8,1 mm	
=_8_006+MCC-WG1	&EFS/23.3	=0+MCC-PE;=_8+MCC-X1	=_8_006+L-H1	.-J	3x	3	1,5 mm²		8,1 mm	
=100+MCC-WG1	&EFS/37.1	=0+MCC-PE;=100+MCC-X2	=100+L-X	.-J	3x	3	1,5 mm²		8,1 mm	
=100+MCC-WG2	&EFS/37.2	=0+MCC-PE;=100+MCC-X2	=100+L-X	.-J	3x	3	1,5 mm²		8,1 mm	
=SEP+MCC-X11.1_WD1	&EFS/22.1	=0+MCC-PE;=SEP+MCC-X11.1	=SEP+JB-X11_PE;=SEP+JB-X11	.-J	4x	4	2,5 mm²		12,5 mm	
=0+PLC-WD1000	&EFS/13.5	=0+MCC-PE;=0+MCC-X1000	=0+PLC-PE;=0+PLC-X1000	.-J	3x	3	1,5 mm²		8,1 mm	
=0+PLC-WG2000	&EFS/13.7	=0+MCC-PE;=0+MCC-X2000	=0+PLC-PE;=0+PLC-X2000	.-J	25x	19	1,5 mm²		24,3 mm	
=0+PLC-WG2001	&EFS/18.3	=0+MCC-X2000;=0+PLC-SC	=0+PLC-X2000	.CY-O	4x	3	0,75 mm²		8,4 mm	
=_9_550+PLC-WG1	&EFS/26.1	=0+PLC-SC;=_9_550+PLC-X2	=_9_550+L-B1	.CY-O	2x	2	0,75 mm²		7,4 mm	
=_7_137+PLC-WG1	&EFS/27.1	=0+PLC-SC;=_7_137+PLC-X2	=_7_137+L-Y1	.CY-O	2x	2	0,75 mm²		7,4 mm	
=_7_131+PLC-WG1	&EFS/28.2	=0+PLC-SC;=_7_131+PLC-X2	=_7_131+L-B1	.CY-O	2x	2	0,75 mm²		7,4 mm	
=100+PLC-WG1	&EFS/36.1	=0+PLC-PE;=100+PLC-X2	=100+L-S1	.-J	3x	3	1,5 mm²		8,1 mm	
=100+PLC-WG2	&EFS/36.2	=0+PLC-PE;=100+PLC-X2	=100+L-S2	.-J	3x	3	1,5 mm²		8,1 mm	
=100+PLC-WG3	&EFS/36.3	=0+PLC-PE;=100+PLC-X2	=100+L-S3	.-J	3x	3	1,5 mm²		8,1 mm	
=100+PLC-WG4	&EFS/36.5	=0+PLC-PE;=100+PLC-X2	=100+L-S4	.-J	3x	3	1,5 mm²		8,1 mm	
=100+PLC-WG5	&EFS/36.6	=0+PLC-PE;=100+PLC-X2	=100+L-S5	.-J	3x	3	1,5 mm²		8,1 mm	

Cable layout

Cable name	/page.path	Source (from)	Target (to)	Cable type	Wires total	Wires used	Cross-section	Length	External diameter	Remark
=100+PLC-WG8	&EFS/37.3	=0+PLC-PE;=100+PLC-X2	=100+L-X	.-J	4x	4	1,5 mm²		8,9 mm	
=100+PLC-WG9	&EFS/37.5	=0+PLC-PE;=100+PLC-X2	=100+L-X	.-J	3x	3	1,5 mm²		8,1 mm	
=100+PLC-WG10	&EFS/37.6	=0+PLC-PE;=100+PLC-X2	=100+L-X	.-J	3x	3	1,5 mm²		8,1 mm	
=100+PLC-WG11	&EFS/36.7	=0+PLC-PE;=100+PLC-X2	=100+L-K1	.-J	3x	3	1,5 mm²		8,1 mm	
=100+PLC-WG12	&EFS/36.8	=0+PLC-PE;=100+PLC-X2	=100+L-K2	.-J	3x	3	1,5 mm²		8,1 mm	
=100+PLC-WG13	&EFS/37.8	=0+PLC-PE;=100+PLC-X2	=100+L-X	.-J	3x	3	1,5 mm²		8,1 mm	
=100+PLC-WG14	&EFS/37.9	=0+PLC-PE;=100+PLC-X2	=100+L-X	.-J	3x	3	1,5 mm²		8,1 mm	
=00+PLC-WF2	&EFS/38.1	=00+PLC-A1	=00+L-X2_DP-BUS	Profibus	1x2x	2	0,64 mm²		8,0 mm	
=SEP+PLC-X10_WG1	&EFS/29.4	=0+PLC-SC;=SEP+PLC-X10	=SEP+JB-X10	.CY-O	2x	2	0,75 mm²		7,4 mm	
=SEP+PLC-X10_WG2	&EFS/30.1	=0+PLC-PE;=SEP+PLC-X10	=SEP+JB-X10_PE;=SEP+JB-X10	.-J	3x	3	1,5 mm²		8,1 mm	
=SEP+PLC-X10_WG3	&EFS/30.5	=0+PLC-SC;=SEP+PLC-X10	=SEP+JB-X10_SC;=SEP+JB-X10	.CY-O	4x	3	0,75 mm²		8,4 mm	
=SEP+PLC-X10_WG5	&EFS/31.1	=0+PLC-PE;=SEP+PLC-X10	=SEP+JB-X10_PE;=SEP+JB-X10	.-J	12x	11	1,5 mm²		17,0 mm	
=SEP+PLC-X11_WG1	&EFS/34.1	=0+PLC-SC;=SEP+PLC-X11	=SEP+JB-X11	.CY-O	8X	8	0,75 mm²		10,2 mm	
=SEP+PLC-X11_WG2	&EFS/33.1	=SEP+PLC-X11	=SEP+JB-X11	.-O	3x	3	0,75 mm²		5,7 mm	
=_3_002+JB-WD1	&EFS/22.1	=SEP+JB-X11;=SEP+JB-X11_PE	=_3_002+L-M1	.-J	4x	4	2,5 mm²		12,5 mm	Westfalia Separator
=_9_022+JB-WG1	&EFS/29.4	=SEP+JB-X10	=_9_022+L-B1	_-O	2x	2	0,5 mm²		4,8 mm	Westfalia Separator
=_9_019+JB-WG1	&EFS/30.1	=SEP+JB-X10;=SEP+JB-X10_PE	=_9_019+L-S1	.-J	3x	3	1,5 mm²		8,1 mm	Westfalia Separator
=_9_123+JB-WG1	&EFS/30.5	=SEP+JB-X10;=SEP+JB-X10_SC	=_9_123+L-S1	.CY-O	4x	4	0,75 mm²		8,4 mm	Westfalia Separator
=_7_024+JB-WG1	&EFS/31.1	=SEP+JB-X10;=SEP+JB-X10_PE	=_7_024+L-M1	.-J	3x	3	1,5 mm²		8,1 mm	Westfalia Separator

Cable layout

Cable name	/page.path	Source (from)	Target (to)	Cable type	Wires total	Wires used	Cross-section	Length	External diameter	Remark
=_7_025+JB-WG1	&EFS/31.2	=SEP+JB-X10;=SEP+JB-X10_PE	=_7_025+L-M1	.-J	3x	3	1,5 mm²		8,1 mm	Westfalia Separator
=_7_022+JB-WG1	&EFS/31.4	=SEP+JB-X10;=SEP+JB-X10_PE	=_7_022+L-M1	.-J	3x	3	1,5 mm²		8,1 mm	Westfalia Separator
=_7_023+JB-WG1	&EFS/31.5	=SEP+JB-X10;=SEP+JB-X10_PE	=_7_023+L-M1	.-J	3x	3	1,5 mm²		8,1 mm	Westfalia Separator
=_7_106+JB-WG1	&EFS/31.7	=SEP+JB-X10;=SEP+JB-X10_PE	=_7_106+L-M1	.-J	3x	3	1,5 mm²		8,1 mm	Westfalia Separator
=_7_034/Y1+JB-WG1	&EFS/32.1	=SEP+JB-X10;=SEP+JB-X10_PE	=_7_034/Y1+L-M1	.-J	3x	3	1,5 mm²		8,1 mm	Westfalia Separator
=_7_032+JB-WG1	&EFS/32.2	=SEP+JB-X10;=SEP+JB-X10_PE	=_7_032+L-M1	.-J	3x	3	1,5 mm²		8,1 mm	Westfalia Separator
=_9_500+JB-WG1	&EFS/33.1	=SEP+JB-X11	=_9_500+L-B1	.-O	4x	4	0,75 mm²		6,2 mm	Westfalia Separator
=_9_018+JB-WG1	&EFS/34.1	=SEP+JB-X11	=_9_018+L-B1	_CY-O	12x	8	0,25 mm²		12,3 mm	Westfalia Separator

0				1				2				3				4				5				6				7				8				9			
			Date	19.06.2018	GEA GEA Westfalia Separator Group GmbH							Benennung/Description: Spare										GEA IO 4							=										
			Editor	HAN/PLE																									+										
			Tested	19.06.2018/GR																																			
Changes	Date	Name			© This document is COPYRIGHTED by GEA Westfalia Separator Group GmbH. All rights being reserved.																	Sprache: Language: en							Zeichnungs Nr.: Drawing No.: 8552-7300-086			DCC & EMB		Bl./Sh. 95 104 Bl./Sh.					



Hinweise zur Installation von Kabeln und Leitungen

Directions for installation from cables and conductors



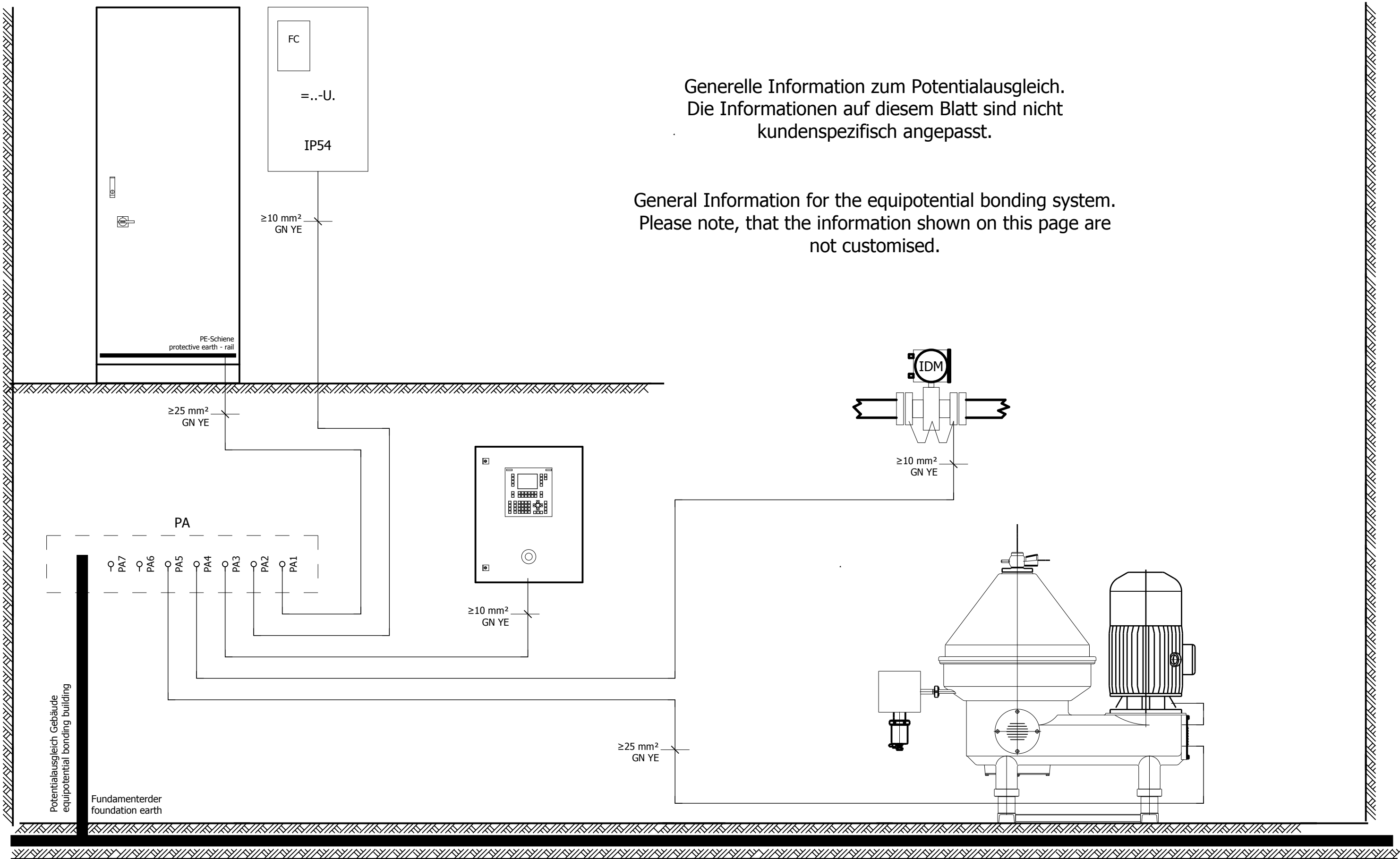
- * Schutzmaßnahmen nach Bestimmungen des VDE bzw. zuständigen EVU ausführen!
All protective measures are to be taken in accordance with electric codes, rules and regulations!
- * Generell sind die Vorgaben des lokalen Energieversorgers zwingend einzuhalten
In general the rules and regulations of the local power authorities have to be followed.
- * Kabel und Leitungen sind so auszuwählen und zu verlegen, dass eine Gefährdung von Personen und der Umgebung ausgeschlossen ist.
Die angegebenen Kabelquerschnitte sind bezogen auf eine Kabellänge von 50m, Kupfer als Leitmaterial, Verlegeart "C" und eine maximale Umgebungstemperatur von 30°C. Siehe hierzu DIN VDE 0298-4 (Verwendung von Kabeln und isolierten Leitungen für Starkstromanlagen - Teil 4: Empfohlene Werte für die Strombelastbarkeit von Kabeln und Leitungen für feste Verlegung in und an Gebäuden und von flexiblen Leitungen).
Neben der Strombelastung sind bei der Errichtung von Kabel- und Leitungsanlagen weitere Faktoren zu berücksichtigen.
Siehe hierzu IEC 60364-5-52 (Errichten von Niederspannungsanlagen - Teil 5: Auswahl und Errichtung elektrischer Betriebsmittel - Kapitel 52: Kabel- und Leitungsanlagen)

Cables and conductors must be selected and installed in a way to eliminate any risk to persons and the environment.
The cable cross-sections specified in these drawings are based on a max. length of 50m, copper as conducting material, laying version "C" and a maximum ambient temperature of 30°C.
Refer also to DIN VDE 0298-4 (Usage of cables and cords in power installations - Part 4: Recommended current-carrying capacity for screened and unscreened cables for fixed wirings in and around buildings and for flexible cables and cords).
Besides the current load additional parameters have to taken into account for the erection of wiring sxstems.
Refer also to IEC 60364-5-52 (Erection of low voltage installations - Part 5: Selection and erection of electrical equipment - Chapter 52: Wiring systems)
- * Messleitungen sind grundsätzlich abgeschirmt zu verlegen!
Measuring lines must always be laid shielded!
- * WD-Kabel getrennt vom WF-/WG-Kabeln verlegen!
WD-cables to be installed separate from WF-/WG-cable!

WD-Kabel	=	Energiekabel < 1kV	WE-Kabel	=	Erdungs-/Potentialausgleichsleiterleiter		
WD-cable		Energy cable < 1kV	WE-cable		Earth-/ Equipotential bonding conductor		
WF-Kabel	=	Datenbus	WG-Kabel	=	Steuer-/ Messkabel	WH-Kabel	=
WF-cable		Data bus	WG-cable		Control-/measuring cable	WH-cable	
							Lichtwellenleiter
							Optical fibre

- * Legende Kabeltypen (Kabelliste)
Legend cable typ (list of cables)

-J	=	Kabel / Leitung mit Schutzleiter Cable / conductor with protective conductor	FC.CY	=	Motorkabel FU mit konzentrischem Außenleiter (PE) Motor cable FC with concentric outer (PE)
-O	=	Kabel / Leitung ohne Schutzleiter Cable / conductor without protective conductor	PL	=	Potentialausgleichsleiter Equipotential bonding conductor
CY-J	=	Kabel / Leitung mit Schutzleiter geschirmt Cable / conductor with protective conductor screened	Ex-CY	=	Steuerleitung blau für eigensichere Stromkreise Control line blue for intrinsically safe circuits
CY-O	=	Kabel / Leitung ohne Schutzleiter geschirmt Cable / conductor without protective conductor screened	CY-TP	=	Datenleitung paarverseilt Data line twisted pair
FC.CY-J	=	Motorkabel FU geschirmt mit Schutzleiter Motor cable FC screened with protective conductor			



Parts list

Function sign	/page.path	GEA part no.	Manufacturer order no..	Name	Manufacturer
=0+MCC-N	&EFS/12.2	9340040	9340040	RiLine60 Sammelschienenhalter (2-polig) bis 800A für E-Cu 12x5 bis 30x10mm Busbar support up to 800A 2-pole for E-Cu 12x5 up to 30x10mm	Rittal
=0+MCC-Q1	&EFS/12.1	283296	283296	Leistungsschalter, 3p, 160A, Ir:125-160A, Ii:1280A Circuit-breaker, 3p, 160A	Eaton-Moeller
=0+MCC-Q1	&EFS/12.1	260166	260166	Drehantrieb, schwarz Rotation drive, black	Eaton-Moeller
=0+MCC-Q1	&EFS/12.1	260191	260191	Verlängerungsachse, 600mm Extension axis	Eaton-Moeller
=0+MCC-Q1	&EFS/12.1	260019	260019	Schraubanschluss, 3p, 1 Schalterseite, Baugrösse 1 Screw connection, 3p, 1 switch side, size 1	Eaton-Moeller
=0+MCC-Q1	&EFS/12.1	Konsole 90X160X100	Konsole 90X200X100	Konsole verzinkt Bracket zined	Zimmermeyer
=0+MCC-F1	&EFS/13.1	0005-1872-220	3RV2011-1DA10	Leistungsschalter, 2,2...3,2A, BGR S00, für den Motorschutz Circuit-breaker, 2,2...3,2A, size S00, motor protection	Siemens
=0+MCC-F2	&EFS/13.2	566532	566532	Sicherungsautomat, D 10A, 1 pol Circuit breaker	AEG
=0+MCC-T1	&EFS/13.1	STU630/2X115	STU630/2X115	Steuerspannungstransformator, Prim.: 210-540V, Sec.: 2x115V, 630VA Control voltage transformer	Block
=0+MCC-X1	&EFS/13.2;	0005-3503-890	1020000000	Durchgangs-Reihenklemme, 2.5 mm², Schraubanschluss, Wemid, dunkelbeige, TS 35 Feed-through terminal, 2,5mm², TS 35	Weidmueller
=0+MCC-X1	&EFS/13.3	1855610000	1855610000	Trenn- und Messtrenn-Reihenklemme, 2.5 mm², Schraubanschluss, Wemid, beige, TS 35 Disconnect and modular disconnect terminal, 2.5mm², TS 35	Weidmueller
=0+MCC-X1000	&EFS/13.5;	0005-3503-890	1020000000	Durchgangs-Reihenklemme, 2.5 mm², Schraubanschluss, Wemid, dunkelbeige, TS 35 Feed-through terminal, 2,5mm², TS 35	Weidmueller
=0+MCC-X2000	&EFS/13.7;	0005-3503-890	1020000000	Durchgangs-Reihenklemme, 2.5 mm², Schraubanschluss, Wemid, dunkelbeige, TS 35 Feed-through terminal, 2,5mm², TS 35	Weidmueller
=0+MCC-F4	&EFS/14.1	0005-1551-030	566513	Sicherungsautomat, C 2A, 1 pol Circuit breaker	AEG
=0+MCC-F5	&EFS/14.2	0005-1551-050	566515	Sicherungsautomat, C 4A, 1 pol Circuit breaker	AEG
=0+MCC-X1.1	&EFS/14.1;	1608570000	1608570000	Federzug-Reihenklemme, 2,5 mm², Wemid, dunkelbeige, TS 35 Feed-through terminal, tension clamp connection	Weidmueller
=0+MCC-X1.2	&EFS/14.1;	1608570000	1608570000	Federzug-Reihenklemme, 2,5 mm², Wemid, dunkelbeige, TS 35 Feed-through terminal, tension clamp connection	Weidmueller
=0+MCC-F6	&EFS/15.1	0005-1551-050	566515	Sicherungsautomat, C 4A, 1 pol Circuit breaker	AEG
=0+MCC-X2.1	&EFS/15.1;	1608570000	1608570000	Federzug-Reihenklemme, 2,5 mm², Wemid, dunkelbeige, TS 35 Feed-through terminal, tension clamp connection	Weidmueller

Parts list

Function sign	/page.path	GEA part no.	Manufacturer order no..	Name	Manufacturer
=0+MCC-X2.2	&EFS/15.1;	1608570000	1608570000	Federzug-Reihenklemme, 2,5 mm², Wemid, dunkelbeige, TS 35 Feed-through terminal, tension clamp connection	Weidmueller
=0+MCC-B1	&EFS/16.2	01141.0-00	01141.0-00	Thermostat für Lüftersteuerung, 1 NO, 0-60°C Thermostat	Ingenieur-Büro Böhm
=0+MCC-M1	&EFS/16.2	3244100	3244100	Filterlüfter, 700/720m³/h 230V, 50/60Hz, RAL 7035 Filter fan, 700/720m³/h 230V, 50/60Hz, RAL 7035	Rittal
=0+MCC-M1	&EFS/16.2	0005-1805-350	3243200	Austrittsfilter für Filterlüfter 3243/3244/3245.xxx RAL 7035 Filter for fan 3243/3244/3245.xxx RAL 7035	Rittal
=0+MCC-K1	&EFS/17.3	0005-4388-140	3SK1111-1AB30	Sicherheitsschaltgerät, Grundgerät Standard 24VAC/DC, 3NO/1NC Emergency stop relay, 24VAC/DC, 3NO/1NC	Siemens
=0+MCC-K3	&EFS/17.8	0005-1870-200	3RH2140-1JB40	Koppelhilfsschütz, 4S 24VDC, mit Diode Coupling contactor relay, 4NO 24VDC, integrated diode	Siemens
=0+MCC-X2	&EFS/17.3	0005-3503-890	1020000000	Durchgangs-Reihenklemme, 2.5 mm², Schraubanschluss, Wemid, dunkelbeige, TS 35 Feed-through terminal, 2,5mm², TS 35	Weidmueller
=_1_001+MCC-T1	&EFS/18.1	0005-5930-160	ATV930D90N4	Frequenzumrichter, 90kW (75kW Phd), 380...480V, 50/60Hz, mit Bremsmodul, IP21 Frequency converter, 90kW (75kW Phd), 380...480V, 50/60Hz, with braking unit, IP21	Groupe Schneider
=_1_001+MCC-K2	&EFS/19.5	3RP2505-1RW30	3RP2505-1RW30	Zeitrelais, Multifunktion, 2 W, 13 Funktionen, 7 Zeitbereiche 0,05 s...100 h, 24...240 Time relay, multi-function, 2 C/O, 13 functions, 7 time set. ranges 0,05 s...100 h, 24... 240	Siemens
=_1_001+MCC-K3	&EFS/19.1	0005-1870-200	3RH2140-1JB40	Koppelhilfsschütz, 4S 24VDC, mit Diode Coupling contactor relay, 4NO 24VDC, integrated diode	Siemens
=_1_001+MCC-K4	&EFS/19.3	0005-1870-200	3RH2140-1JB40	Koppelhilfsschütz, 4S 24VDC, mit Diode Coupling contactor relay, 4NO 24VDC, integrated diode	Siemens
=_1_001+MCC-K1	&EFS/20.3	0005-1870-200	3RH2140-1JB40	Koppelhilfsschütz, 4S 24VDC, mit Diode Coupling contactor relay, 4NO 24VDC, integrated diode	Siemens
=_1_001+MCC-P1	&EFS/21.1	0005-1187-040	BW70.29-230V/50-60Hz	Stundenzähler, auf Montageplatte, 230V/50-60Hz Hours counter	Hugo Mueller Elektrotechnik
=_3_002+MCC-F1	&EFS/22.1	0005-1872-070	3RV2011-0FA10	Leistungsschalter, 0,35...0,5A, BGR S00, für den Motorschutz Circuit-breaker, 0,35...0,5A, size S00, motor protection	Siemens
=_3_002+MCC-F1	&EFS/22.1	3RV2901-1E	3RV2901-1E	Hilfsschalter, frontseitig querliegend 1S+1OE Auxiliary switch, transverse on the front 1NO + 1NC	Siemens
=_3_002+MCC-K1	&EFS/22.8	0005-4033-040	2966171	Relais Einzelkontakt, PLC-Interface, Miniaturrelais mit Leistungskontakt, 1 Wechsler, Un=24 Relay single contact	Phoenix Contact
=_3_002+MCC-Q1	&EFS/22.4	0005-1871-120	3RT2016-1AP01	Schütz, AC-3, 4KW/400V, 1S, AC 230V, 50/60 Hz Contactor, AC-3, 4KW/400V, 1NO, AC 230V, 50/60 Hz	Siemens
=_3_002+MCC-Q1	&EFS/22.4	3RT2916-1CD00	3RT2916-1CD00	RC-Glied, AC 127...240V, DC 150...250V RC-element, AC 127...240 V, DC 150...250V, surge suppressor, for mounting onto	Siemens
=SEP+MCC-X11.1	&EFS/22.1	0005-3503-890	1020000000	Durchgangs-Reihenklemme, 2.5 mm², Schraubanschluss, Wemid, dunkelbeige, TS 35 Feed-through terminal, 2,5mm², TS 35	Weidmueller

Parts list

Function sign	/page.path	GEA part no.	Manufacturer order no..	Name	Manufacturer
=_8+MCC-F1	&EFS/23.1	0005-1551-040	566514	Sicherungsautomat, C 3A, 1 pol Circuit breaker	AEG
=_8+MCC-K1	&EFS/23.8	0005-4033-040	2966171	Relais Einzelkontakt, PLC-Interface, Miniaturelais mit Leistungskontakt, 1 Wechsler, Un=24 Relay single contact	Phoenix Contact
=_8+MCC-T1	&EFS/23.1	ST63/23/24	ST63/23/24	Steuerspannungstransformator, Prim.: 230V, Sec.: 24V, 63VA Control voltage transformer	Block
=_8+MCC-X1	&EFS/23.1;	0005-3503-890	1020000000	Durchgangs-Reihenklemme, 2.5 mm², Schraubanschluss, Wemid, dunkelbeige, TS 35 Feed-through terminal, 2,5mm², TS 35	Weidmueller
=_8+MCC-X1	&EFS/23.4	1855610000	1855610000	Trenn- und Messtrenn-Reihenklemme, 2.5 mm², Schraubanschluss, Wemid, beige, TS 35 Disconnect and modular disconnect terminal, 2.5mm², TS 35	Weidmueller
=100+MCC-X2	&EFS/37.1;	0005-3503-890	1020000000	Durchgangs-Reihenklemme, 2.5 mm², Schraubanschluss, Wemid, dunkelbeige, TS 35 Feed-through terminal, 2,5mm², TS 35	Weidmueller
+MCC-CABINET	&ELU/48.1	8805500	8805500	Topschrank-System lackiert RAL 7035, mit Montageplatte eintürig Top enclosure system painted RAL 7035, with mounting plate single-door	Rittal
+MCC-CABINET	&ELU/48.1	8105235	8105235	Seitenwand für Schrank (HxT): 2000x500mm, RAL 7035 Side panels, for 2000 X 500 mm RAL 7035	Rittal
+MCC-FITTINGS	&ELU/48.0	8556-9999-000	KLEINMATERIAL	Kleinmaterial Schaltschrank Incidentals control cabinet	Plenge
+MCC-FITTINGS	&ELU/48.0	4118000	4118000	Schaltplantaschen aus Stahlblech für Türbreite 800mm Wiring plan pockets of sheet steel for 800 mm wide doors	Rittal
+MCC-FITTINGS	&ELU/48.0	4192000	4192000	Kabelabfangschiene für 800mm Schrankbreite Cable clamp rail for 800 mm enclosure width	Rittal
+MCC-PLINTH	&ELU/48.1	8601800	8601800	Sockel-Element, vorne und hinten, BxH 800x100mm Base/plinth component, front and rear, WxH 800x100mm	Rittal
+MCC-PLINTH	&ELU/48.1	8601050	8601050	Sockel-Blende, seitlich, BxH 500x100mm Base/plinth trim, side, WxH 500x200mm	Rittal

Parts list

Function sign	/page.path	GEA part no.	Manufacturer order no..	Name	Manufacturer
=0+PLC-F2	&EFS/13.5	0005-1551-060	566516	Sicherungsautomat, C 6A, 1 pol Circuit breaker	AEG
=0+PLC-SC	&EFS/13.7;	1600480000	1600480000	Klemmbügel für Schirmanschluss, zur Montage auf Sammelschiene 10x3mm Retaining clip for shield connection	Weidmueller
=0+PLC-T2	&EFS/13.5	0005-1407-050	6EP1333-2BA20-0AA0	Einphasennetzgerät, Prim.: 120/230V, 50/60Hz, Sec.: 24VDC, 5A Single phase power supply, Prim.: 120/230V, 50/60Hz, Sec.: 24VDC, 5A	Siemens
=0+PLC-X2	&EFS/13.5;	0005-3503-890	1020000000	Durchgangs-Reihenklemme, 2.5 mm², Schraubanschluss, Wemid, dunkelbeige, TS 35 Feed-through terminal, 2,5mm², TS 35	Weidmueller
=0+PLC-X2	&EFS/13.6	1855610000	1855610000	Trenn- und Messtrenn-Reihenklemme, 2.5 mm², Schraubanschluss, Wemid, beige, TS 35 Disconnect and modular disconnect terminal, 2.5mm², TS 35	Weidmueller
=0+PLC-X1000	&EFS/13.5;	0005-3503-890	1020000000	Durchgangs-Reihenklemme, 2.5 mm², Schraubanschluss, Wemid, dunkelbeige, TS 35 Feed-through terminal, 2,5mm², TS 35	Weidmueller
=0+PLC-X2000	&EFS/13.7;	0005-3503-890	1020000000	Durchgangs-Reihenklemme, 2.5 mm², Schraubanschluss, Wemid, dunkelbeige, TS 35 Feed-through terminal, 2,5mm², TS 35	Weidmueller
=0+PLC-F7	&EFS/15.5	0005-1551-050	566515	Sicherungsautomat, C 4A, 1 pol Circuit breaker	AEG
=0+PLC-X2.1	&EFS/15.5..	1608570000	1608570000	Federzug-Reihenklemme, 2,5 mm², Wemid, dunkelbeige, TS 35 Feed-through terminal, tension clamp connection	Weidmueller
=0+PLC-X2.2	&EFS/15.5..	1608570000	1608570000	Federzug-Reihenklemme, 2,5 mm², Wemid, dunkelbeige, TS 35 Feed-through terminal, tension clamp connection	Weidmueller
=0+PLC-S0	&EFS/17.1	3SB3000-1HA20	3SB3000-1HA20	PILZDRUCKTASTER RUND 40MM MUSHROOM-SHAPED PUSHBUTTON	Siemens
=0+PLC-S0	&EFS/17.1	3SB3400-0M	3SB3400-0M	Schaltelement mit 1 Schaltglied, 1OE, Montageueberwachungskontakt Contact bloc with 1 contact 1NC	Siemens
=0+PLC-S0	&EFS/17.1	0005-4393-000	3SB3921-0AB	GELBES UNTERLEGSCHILD, RUND SELBSTKLEBEND, OHNE AUFSCHRIFT 80MM YELLOW BACKING PLATE, ROUND SELF-ADHESIVE WITHOUT INSCRIPTION 80MM	Siemens
=0+PLC-SC1	&EFS/18.3;	1600480000	1600480000	Klemmbügel für Schirmanschluss, zur Montage auf Sammelschiene 10x3mm Retaining clip for shield connection	Weidmueller
=_9_034+PLC-X2	&EFS/24.1	0005-3503-890	1020000000	Durchgangs-Reihenklemme, 2.5 mm², Schraubanschluss, Wemid, dunkelbeige, TS 35 Feed-through terminal, 2,5mm², TS 35	Weidmueller
=Y+PLC-X2	&EFS/25.1;	0005-4705-000	1041100000	Durchgangs-Doppelklemme, 2.5 mm², Schraubanschluss, Wemid, dunkelbeige, TS 35 Feed-through terminal	Weidmueller
=_9_550+PLC-X2	&EFS/26.1	0005-3503-890	1020000000	Durchgangs-Reihenklemme, 2.5 mm², Schraubanschluss, Wemid, dunkelbeige, TS 35 Feed-through terminal, 2,5mm², TS 35	Weidmueller
=_7_137+PLC-X2	&EFS/27.1;	0005-3503-890	1020000000	Durchgangs-Reihenklemme, 2.5 mm², Schraubanschluss, Wemid, dunkelbeige, TS 35 Feed-through terminal, 2,5mm², TS 35	Weidmueller
=_7_131+PLC-T1	&EFS/28.1	KFD2-SCD-Ex1.LK	KFD2-SCD-Ex1.LK	Trennwandler SMART Current Driver	Pepperl + Fuchs

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=_7_131+PLC-X2	&EFS/28.2	0005-3503-890	1020000000	Durchgangs-Reihenklemme, 2.5 mm², Schraubanschluss, Wemid, dunkelbeige, TS 35 Feed-through terminal, 2,5mm², TS 35	Weidmueller
=_9_022+PLC-T1	&EFS/29.2	0005-1487-010	KFD2-UFC-1.D	Drehzahlmessgerät, 24VDC, Eingangsfrequenz 1 mHz ... 12kHz, Stromausgang 0/4mA ... Speed measuring unit	Pepperl + Fuchs
=SEP+PLC-X10	&EFS/29.4;	0005-3503-890	1020000000	Durchgangs-Reihenklemme, 2.5 mm², Schraubanschluss, Wemid, dunkelbeige, TS 35 Feed-through terminal, 2,5mm², TS 35	Weidmueller
=SEP+PLC-X11	&EFS/33.1;	0005-3503-890	1020000000	Durchgangs-Reihenklemme, 2.5 mm², Schraubanschluss, Wemid, dunkelbeige, TS 35 Feed-through terminal, 2,5mm², TS 35	Weidmueller
=0+PLC-SC	&EFS/34.5	1592810000	1592810000	Klemmbügel für Schirmanschluss, zur Montage auf Sammelschiene 10x3mm Retaining clip for shield connection	Weidmueller
=_9_018+PLC-K1	&EFS/35.1	0005-4033-040	2966171	Relais Einzelkontakt, PLC-Interface, Miniaturrelais mit Leistungskontakt, 1 Wechsler, Un=24 Relay single contact	Phoenix Contact
=100+PLC-X2	&EFS/36.1..	0005-4705-000	1041100000	Durchgangs-Doppelklemme, 2.5 mm², Schraubanschluss, Wemid, dunkelbeige, TS 35 Feed-through terminal	Weidmueller
=100+PLC-K1	&EFS/37.3	0005-4033-040	2966171	Relais Einzelkontakt, PLC-Interface, Miniaturrelais mit Leistungskontakt, 1 Wechsler, Un=24 Relay single contact	Phoenix Contact
=100+PLC-K2	&EFS/37.5	0005-4033-040	2966171	Relais Einzelkontakt, PLC-Interface, Miniaturrelais mit Leistungskontakt, 1 Wechsler, Un=24 Relay single contact	Phoenix Contact
=100+PLC-K3	&EFS/37.6	0005-4033-040	2966171	Relais Einzelkontakt, PLC-Interface, Miniaturrelais mit Leistungskontakt, 1 Wechsler, Un=24 Relay single contact	Phoenix Contact
=100+PLC-K4	&EFS/37.8	0005-4033-040	2966171	Relais Einzelkontakt, PLC-Interface, Miniaturrelais mit Leistungskontakt, 1 Wechsler, Un=24 Relay single contact	Phoenix Contact
=100+PLC-K5	&EFS/37.9	0005-4033-040	2966171	Relais Einzelkontakt, PLC-Interface, Miniaturrelais mit Leistungskontakt, 1 Wechsler, Un=24 Relay single contact	Phoenix Contact
=00+PLC-A1	&EFS/38.1	0005-4050-934	AB7652	Anybus X-Gateway, Profibus-Slave / Profinet IO Device (Slave) Anybus X-Gateway, Profibus-Slave / Profinet IO Device (Slave)	HMS Industrial Networks
=00+PLC-A1	&EFS/38.1	Patchkabel CAT6 S/FTP 1,0m grün	Patchkabel CAT6 S/FTP 1,0m grün	Patchkabel CAT6 S/FTP 1,0m grün Patchkabel CAT6 S/FTP 1,0m green	MPC
=00+PLC-A1	&EFS/38.1	700-972-7BB50	700-972-7BB50	Profi-Busanschlussstecker, mit PG Anschluss, 90° Process bus connector	Helmholz
=00+PLC-P1	&EFS/41.1	0005-4760-010	6AV2124-5DC00-0AC0	Touchpanel KTP400 Comfort, 4" Farb-Display, 24VDC Touchpanel KTP400 Comfort, 4" Color-Display, 24VDC	Siemens
=00+PLC-P1	&EFS/41.1	0005-3668-100	6AV2181-8XP00-0AX0	SIMATIC HMI SD-Speicherkarte 2 GB SIMATIC HMI SD memory card 2 GB	Siemens
=00+PLC-K1	&EFS/42.1	0005-4760-600	6ES7215-1AL40-0XB0	Zentralbaugruppe, SIMATIC S7-1200, CPU 1215C, 14DI/10DO/2AI/2AO CPU module	Siemens
=00+PLC-K1	&EFS/42.1	0005-4278-060	6XV1850-2GH20	SIMATIC NET, IND. ETHERNET TP CORD RJ45/RJ45, LAENGE 2 M Simatic NET, ind. ethernet TP cord RJ45/RJ45, length 2m	Siemens

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=00+PLC-K1	&EFS/42.1	6ES7954-8LE02-0AA0	6ES7954-8LE02-0AA0	Speicherkarte, 12 MByte, MMC Memory card, 12 MByte, MMC	Siemens
=00+PLC-K2	&EFS/42.4	0005-4281-100	6ES7234-4HE32-0XB0	Analogein-/ausgabe, SM 1234, 4AE/2AA, U/I, 13/14BIT Analoge module in-/output	Siemens
=00+PLC-K3	&EFS/42.5	0005-4280-040	6ES7221-1BH32-0XB0	Digital Eingangsmodul, SM 1221, 16DI Digital input, SM 1221, 16DI	Siemens
=00+PLC-K4	&EFS/42.6	0005-4280-060	6ES7222-1BH32-0XB0	Digital Ausgangsmodul, SM 1222, 16 DO, 24VDC/0,5A Digital output, SM 1222, 16 DO, 24VDC/0,5A	Siemens
=00+PLC-K5	&EFS/42.8	0005-4281-100	6ES7234-4HE32-0XB0	Analogein-/ausgabe, SM 1234, 4AE/2AA, U/I, 13/14BIT Analoge module in-/output	Siemens
+PLC-Blindstopfen	&ELU/50.2	0005-8001-580	53100006	Dichteinsatz 6x 14 Sealing insert 6x 14	Lapp
+PLC-Blindstopfen	&ELU/50.2	WJ-DM 32/4x8	WJ-DM 32/4x8	Mehrfacheinsatz für M32, 4x 8mm Kabel-ø Multiple sealing insert for M32	Jacob
+PLC-Blindstopfen	&ELU/50.2	0005-8001-560	53100008	Dichteinsatz 8x 14 Sealing insert 8x 14	Lapp
+PLC-CABINET	&ELU/50.1	0005-0059-070	1012600	Kompakt-Schaltschrank AE Edelstahl 1.4301, mit Montageplatte Compact enclosure AE stainless steel 1.4301, with mounting plate	Rittal
+PLC-FITTINGS	&ELU/50.0	8556-9999-000	KLEINMATERIAL	Kleinmaterial Schaltschrank Incidentals control cabinet	Plenge
+PLC-FITTINGS	&ELU/50.0	1080100000	1080100000	Endwinkel (Schirmschiene einseitig) End angle	Weidmueller
+PLC-FITTINGS	&ELU/50.0	Abdeckung-OP	Abdeckung-OP	Abdeckung-OP für Tropfwasserschutz , 135x300x3mm Cover OP for dripping water protection	Plenge
+PLC-Verschraubung	&ELU/50.1	0005-4487-930	70-20-060-000-13-041	Verschraubung Polyamid, M20x1,5, RAL 7032, halogenfrei, d=8-13mm Screw connection polyamide, M20x1,5, RAL 7032, halogen free, 8-13mm	Jacob
+PLC-Verschraubung	&ELU/50.1	0005-4443-900	81-20-866-0000-04	Blindstopfen Polyamid, M20x1,5, RAL 7035 Blanking plug polyamide, M20x1,5, RAL 7035	Jacob
+PLC-Verschraubung	&ELU/50.1	0013-0131-630	82-20-02-0-04	Gegenmutter Messing vernickelt, M20x1,5 Counter nut brass nickeled, M20x1,5	Jacob
+PLC-Verschraubung	&ELU/50.1	0005-4488-930	70-25-060-000-17-041	Verschraubung Polyamid, M25x1,5, RAL 7032, halogenfrei, d=11-17mm Screw connection polyamide, M25x1,5, RAL 7032, halogen free, 11-17mm	Jacob
+PLC-Verschraubung	&ELU/50.1	0013-0132-630	82-25-02-0-04	Gegenmutter Messing vernickelt, M25x1,5 Counter nut brass nickeled, M25x1,5	Jacob
+PLC-Verschraubung	&ELU/50.1	0005-4489-930	70-32-060-000-21-041	Verschraubung Polyamid, M32x1,5, RAL 7032, halogenfrei, d=15-21mm Screw connection polyamide, M32x1,5, RAL 7032, halogen free, 15-21mm	Jacob
+PLC-Verschraubung	&ELU/50.1	0013-0133-630	82-32-02-0-04	Gegenmutter Messing vernickelt, M32x1,5 Counter nut brass nickeled, M32x1,5	Jacob

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Function sign	/page.path	GEA part no.	Manufacturer order no..	Name	Manufacturer
+PLC-Verschraubung	&ELU/50.1	WJ-DM 32/6x6	WJ-DM 32/6x6	Mehrfacheinsatz für M32, 6x 6mm Kabel-ø Multiple sealing insert for M32	Jacob
+PLC-Verschraubung	&ELU/50.1	WJ-DM 32/4x8	WJ-DM 32/4x8	Mehrfacheinsatz für M32, 4x 8mm Kabel-ø Multiple sealing insert for M32	Jacob
+PLC-Verschraubung	&ELU/50.1	0005-4490-930	70-40-060-000-28-041	Verschraubung Polyamid, M40x1,5, RAL 7032, halogenfrei, d=19-28mm Screw connection polyamide, M40x1,5, RAL 7032, halogen free, 19-28mm	Jacob
+PLC-Verschraubung	&ELU/50.1	0013-0134-630	82-40-02-0-04	Gegenmutter Messing vernickelt, M40x1,5 Counter nut brass nickeled, M40x1,5	Jacob